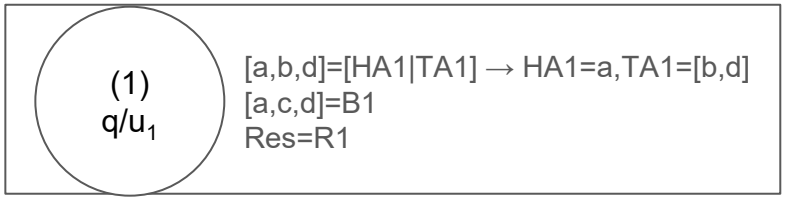
Seminary #2

Union predicate

```
[u1]    union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2]    union([HA|TA], B, [HA|R]):-
                %not(member(HA,B)),
[u21]           union(TA, B, R).
[u3]    union([], B, B).
```

Union predicate

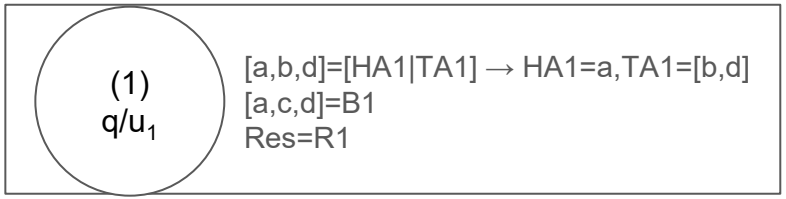



```
[m1] member(a, [a,c,d]).
```

C1: q/head has unified

```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```

Union predicate

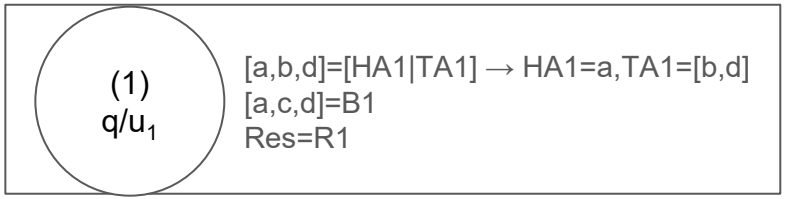



```
[m1] member(a, [a,c,d]).
```

C1: q/head has unified
C2: fact? NO

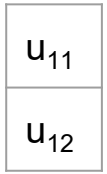
```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```

Union predicate




```
[m1] member(a, [a,c,d]).
```

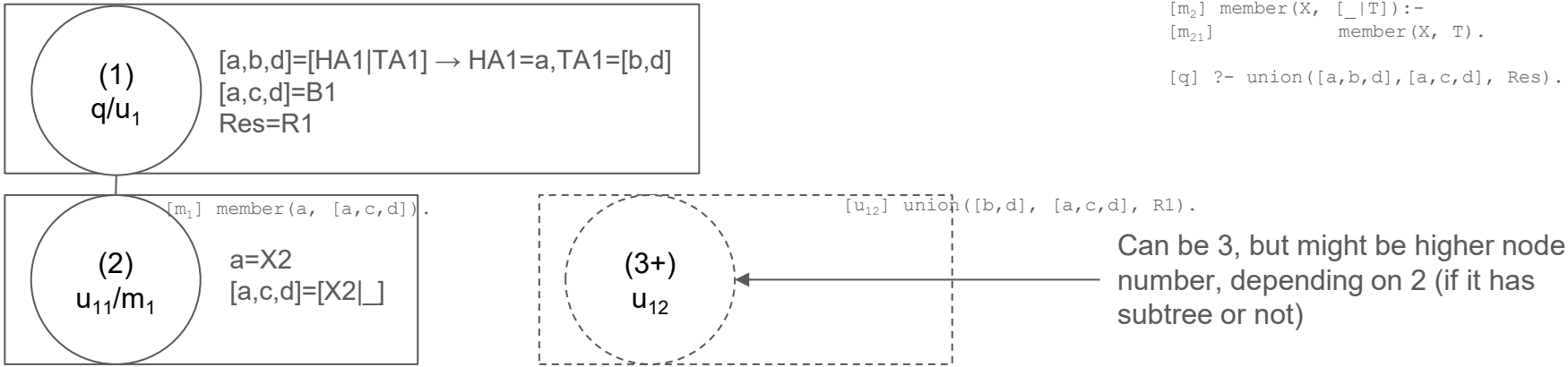
Stack




```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```

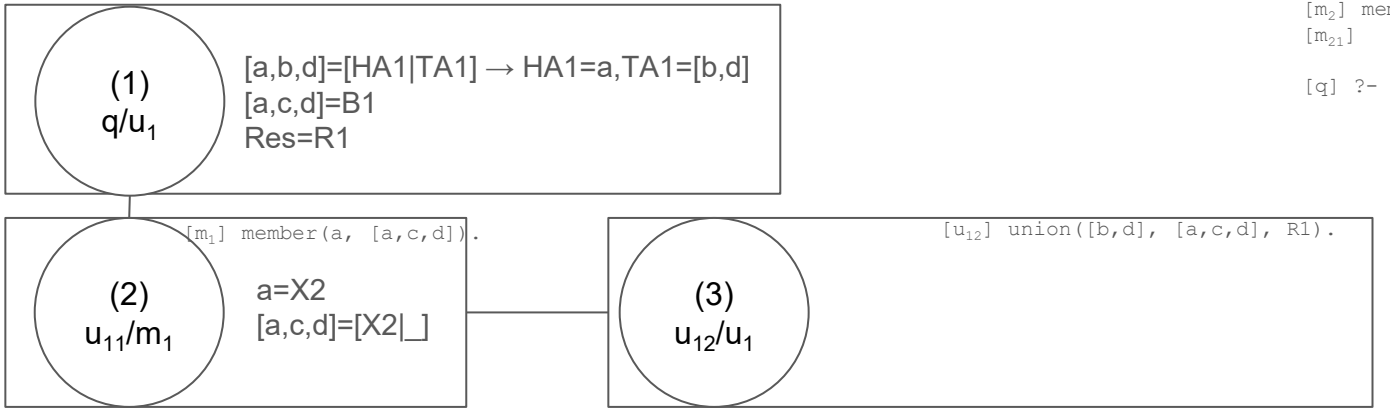
Union predicate

```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```



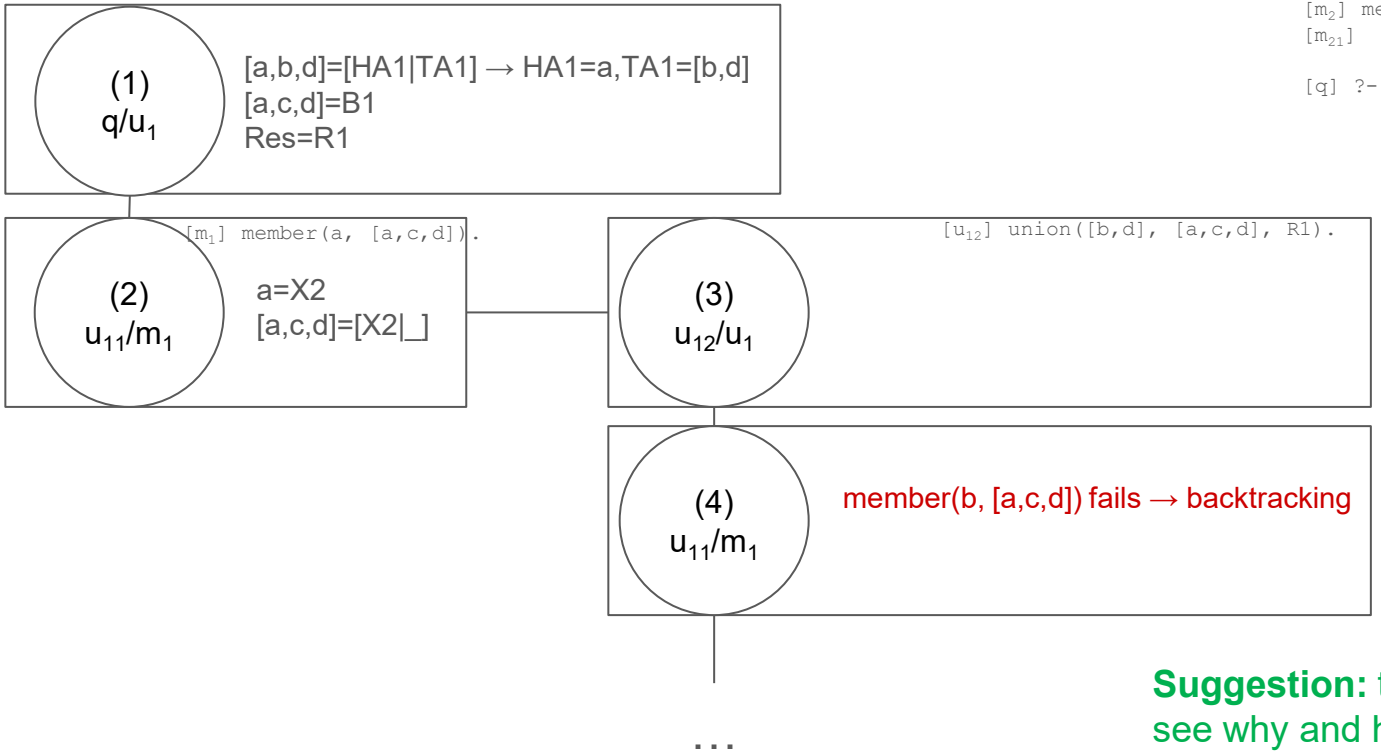
Union predicate

```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```



Union predicate

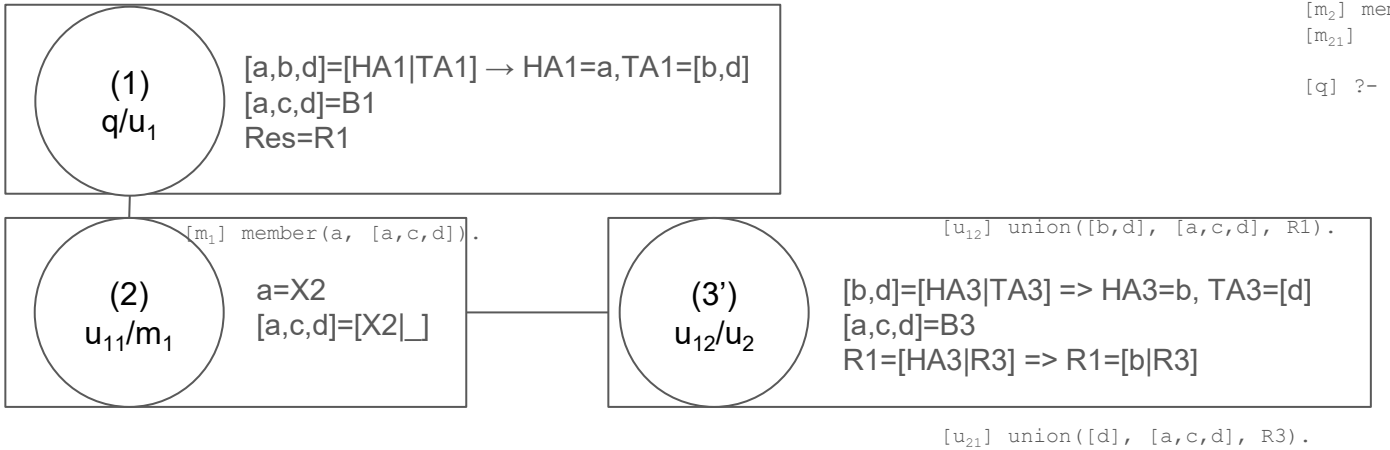
```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```



Suggestion: trace it yourselves and see why and how it fails

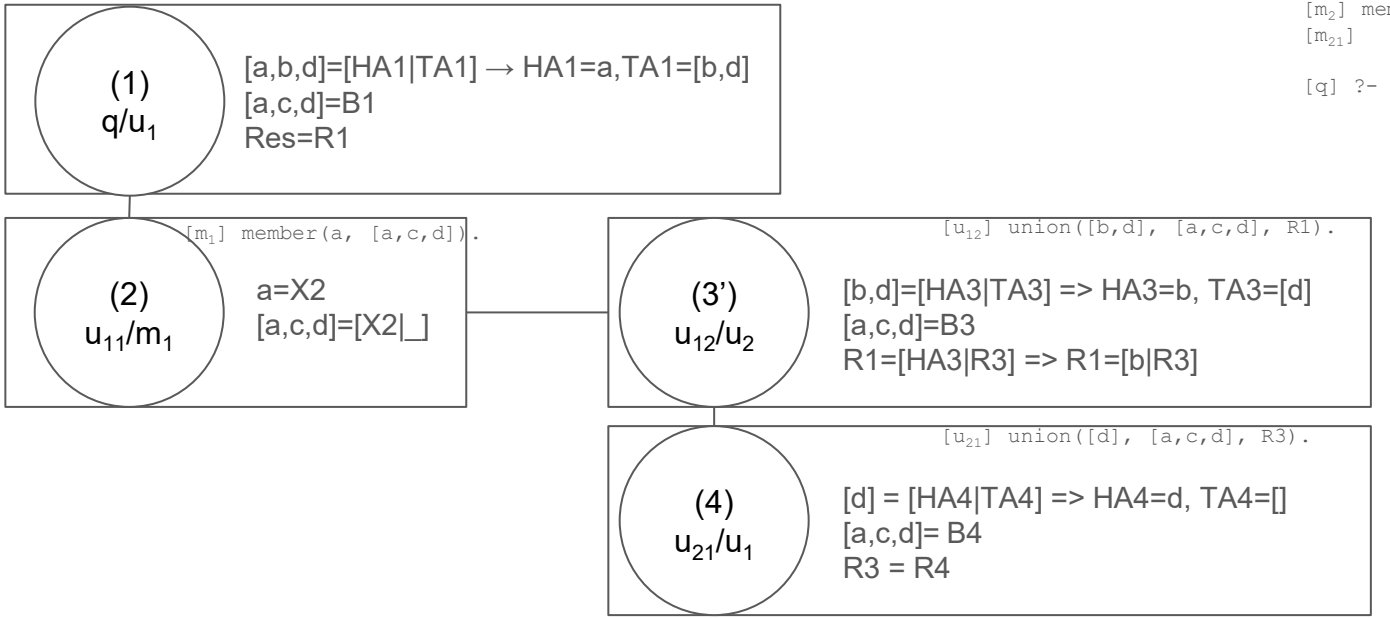
Union predicate

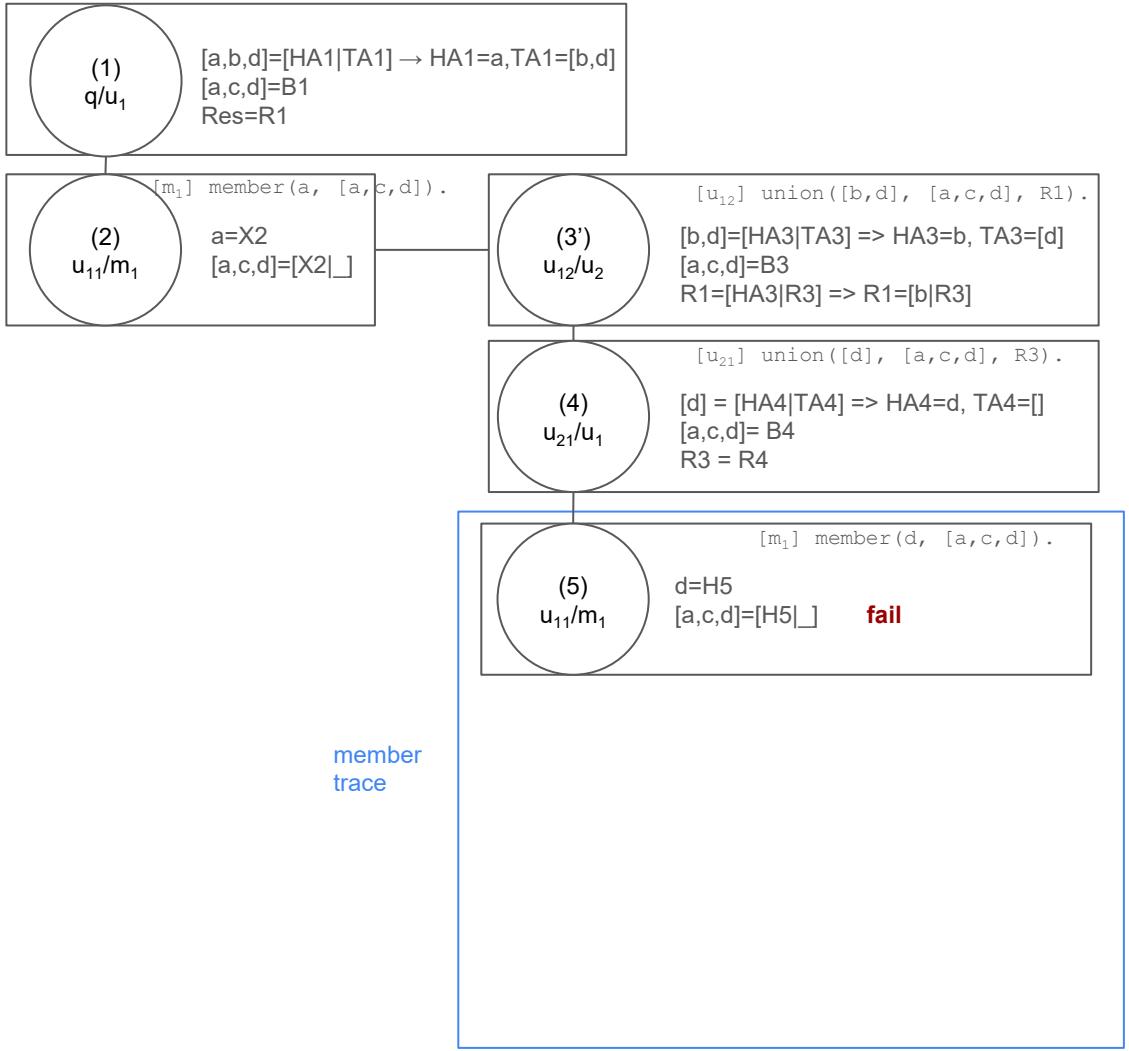
```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```



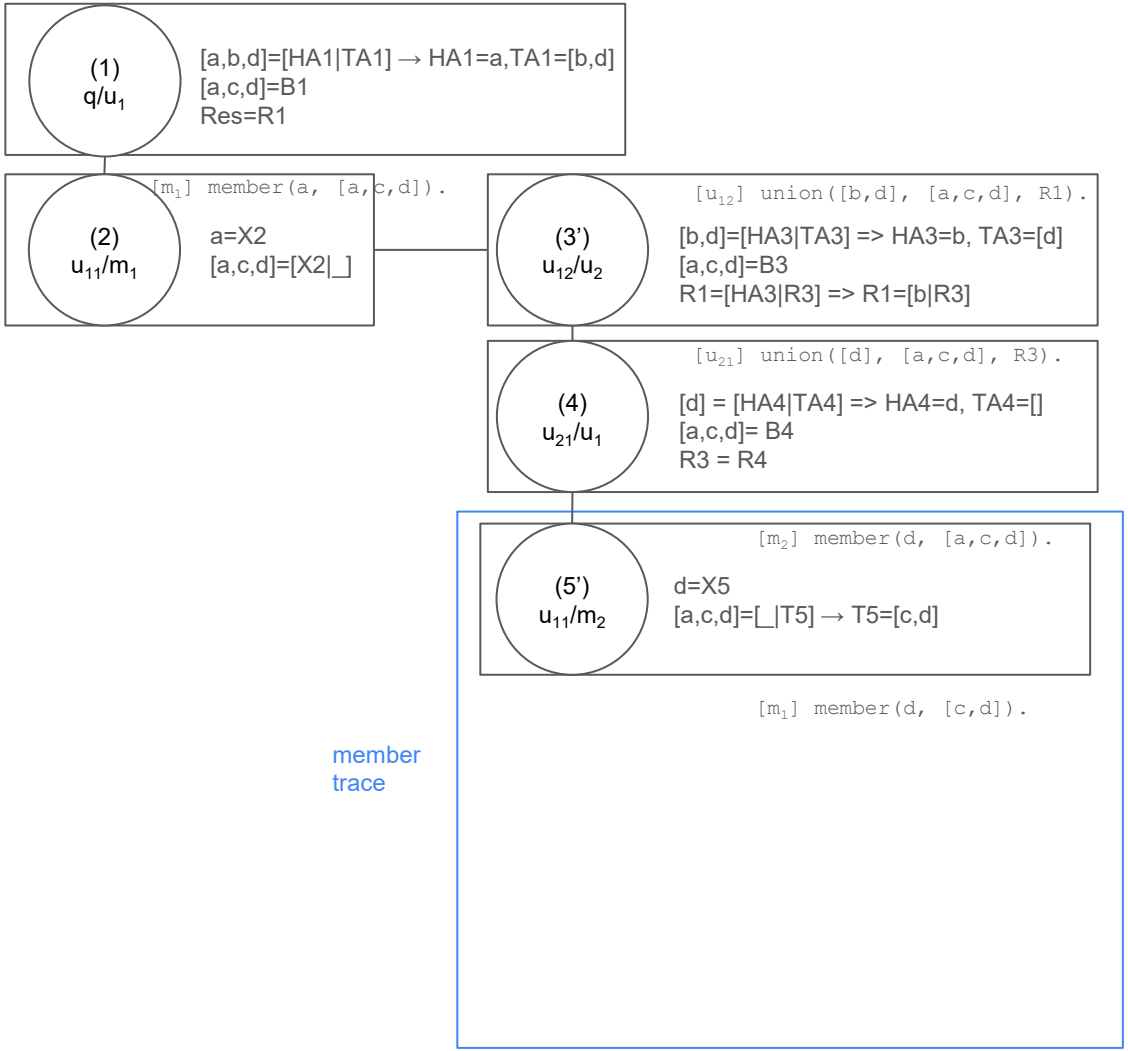
Union predicate

```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```

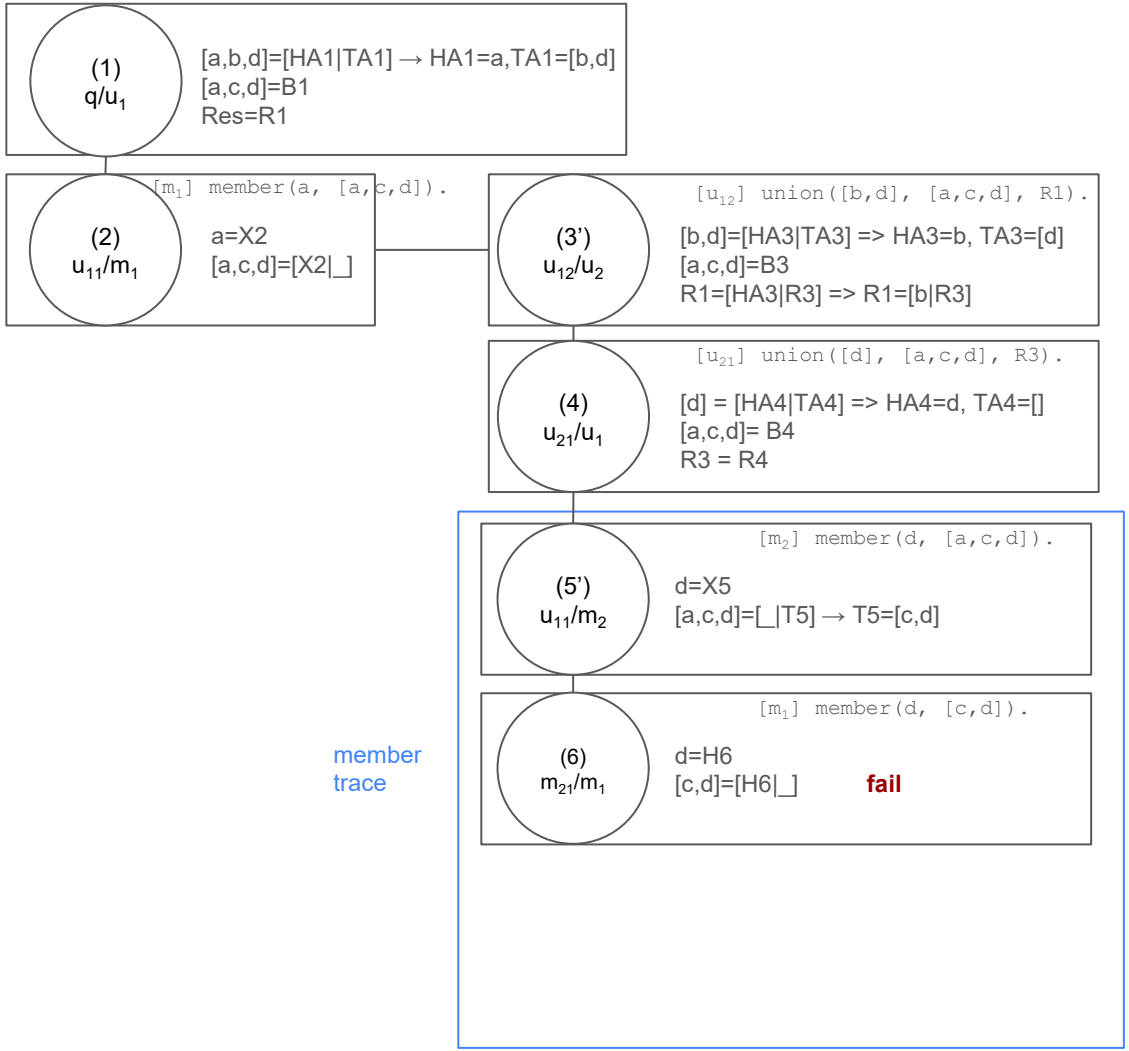




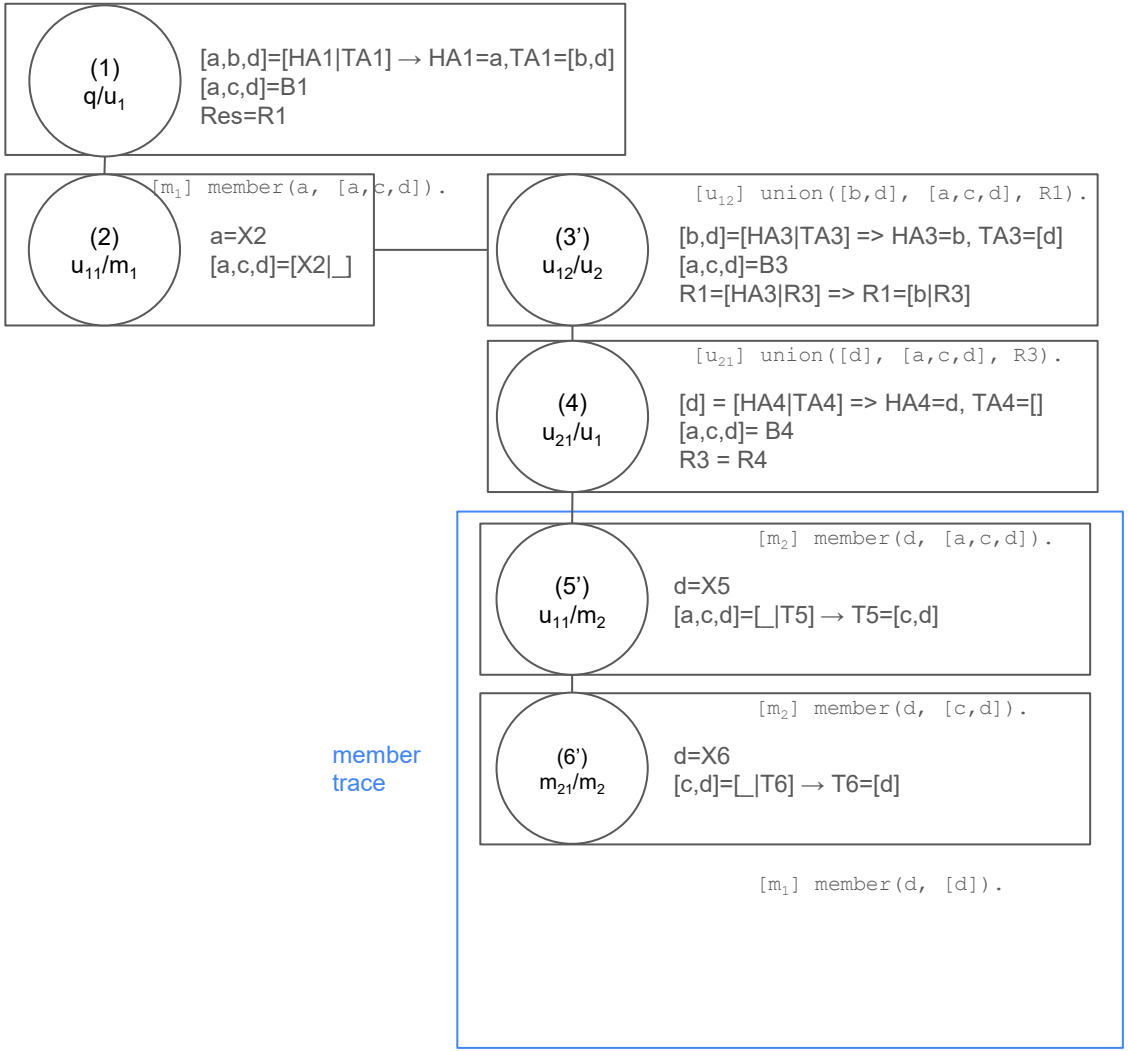

```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```



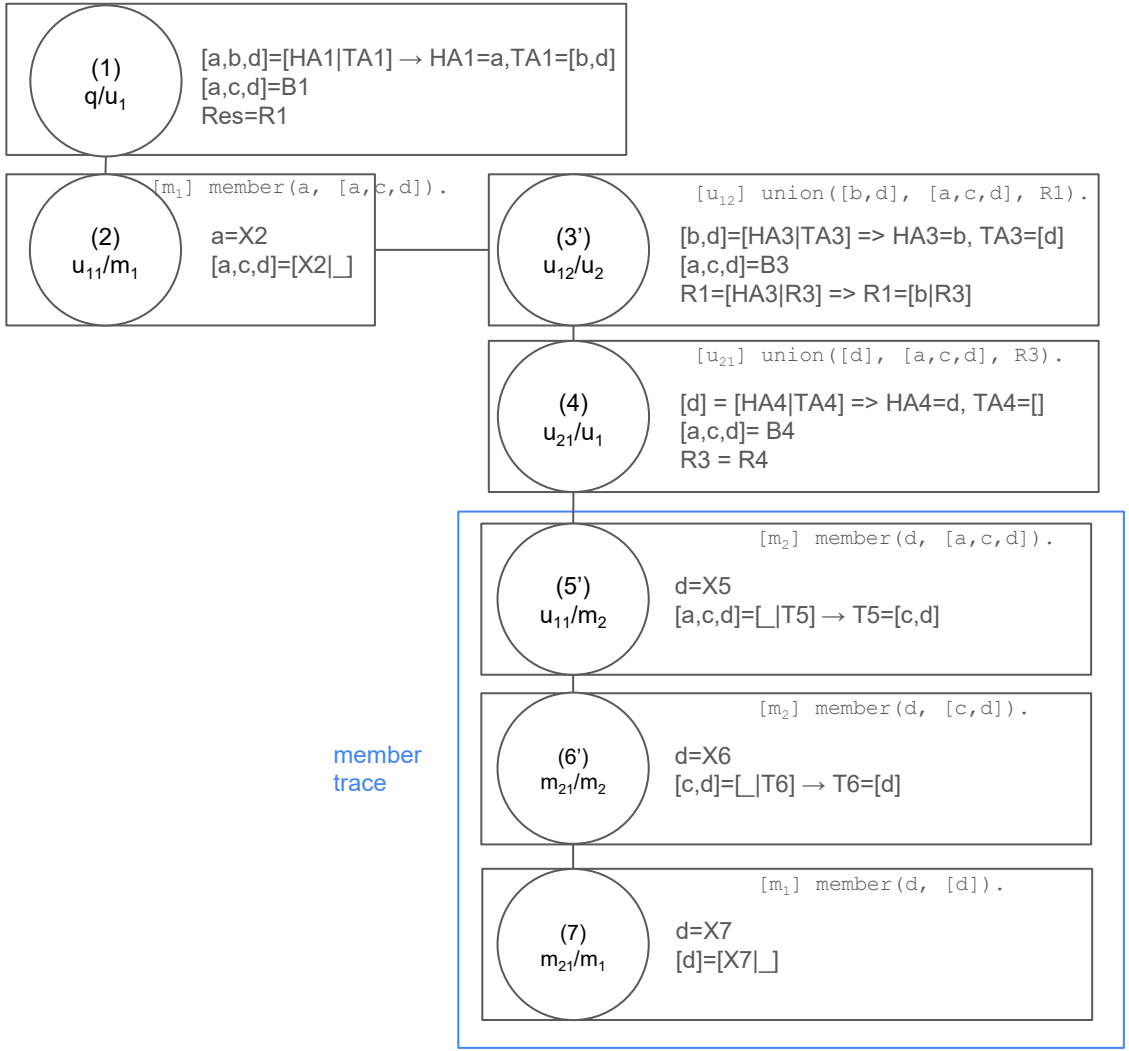

```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```



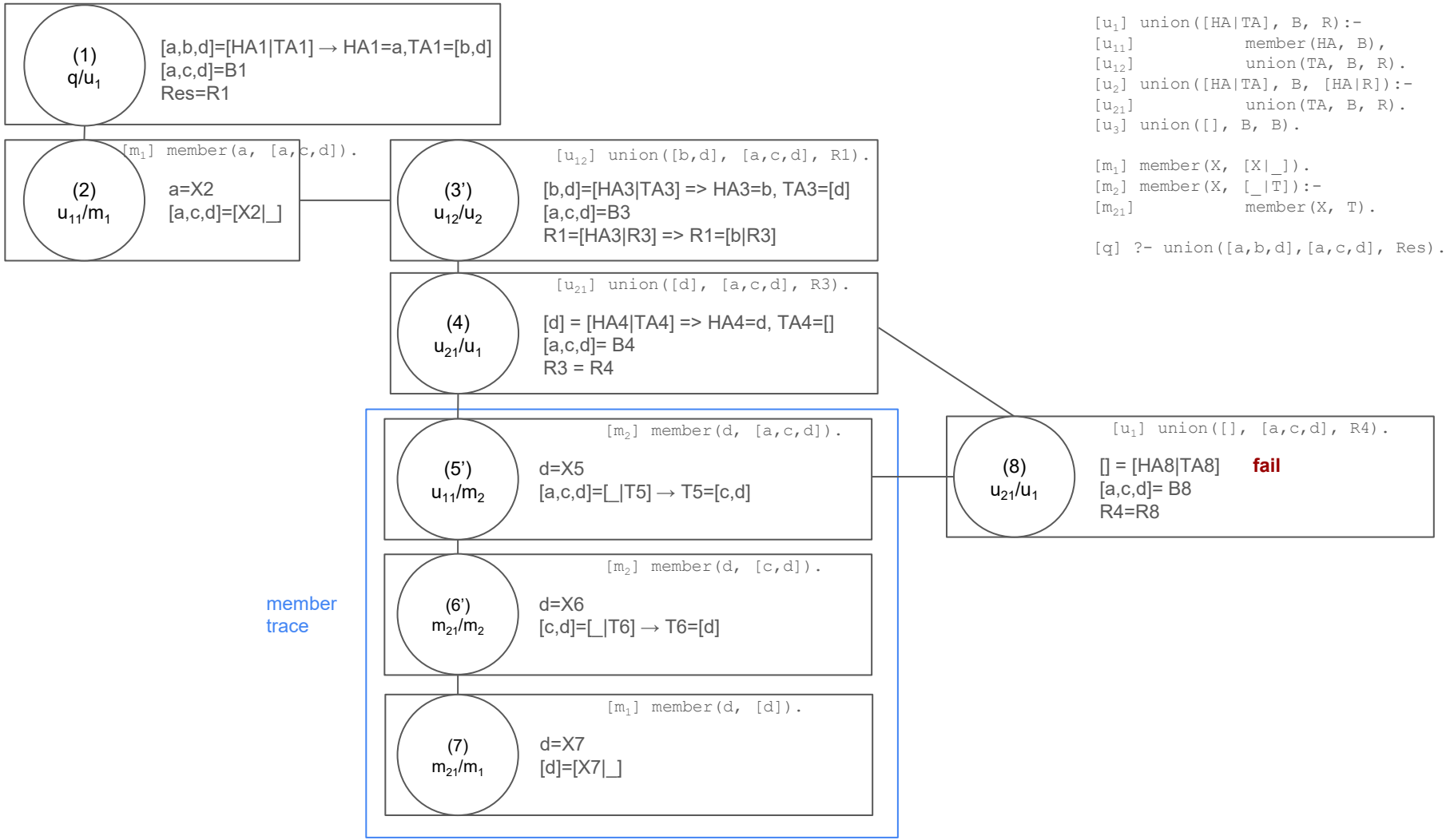

```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```

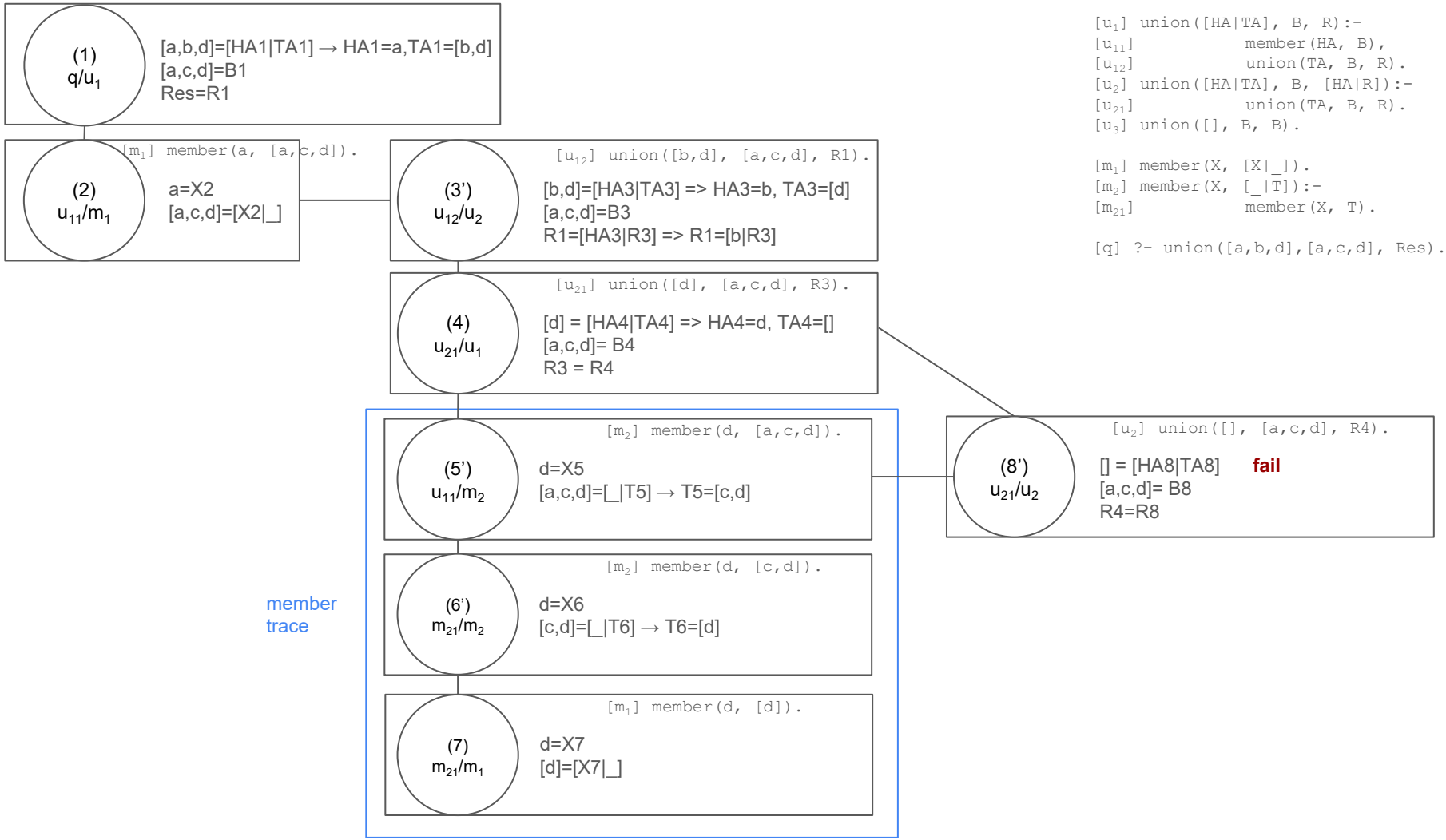


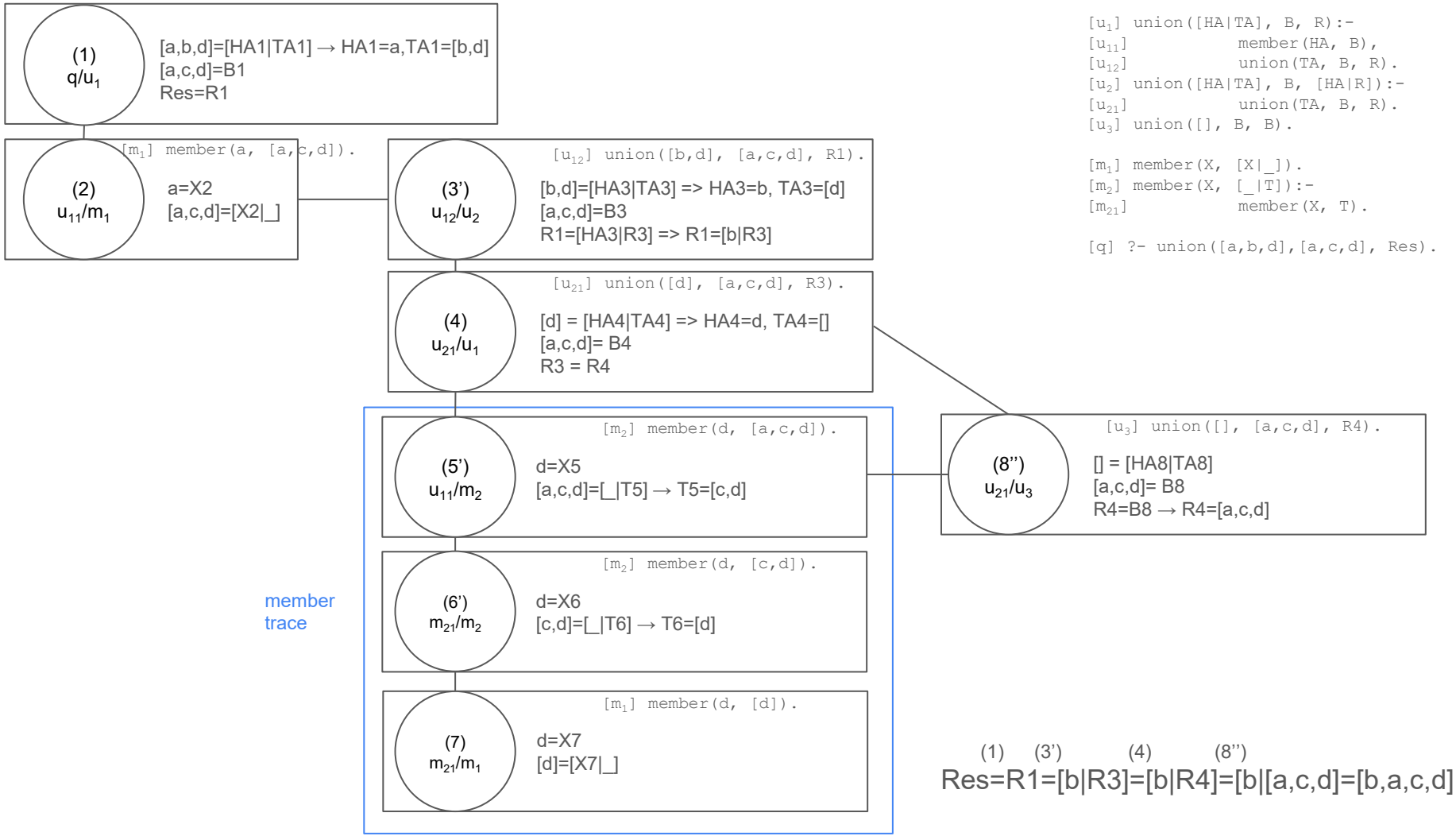

```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```




```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
```

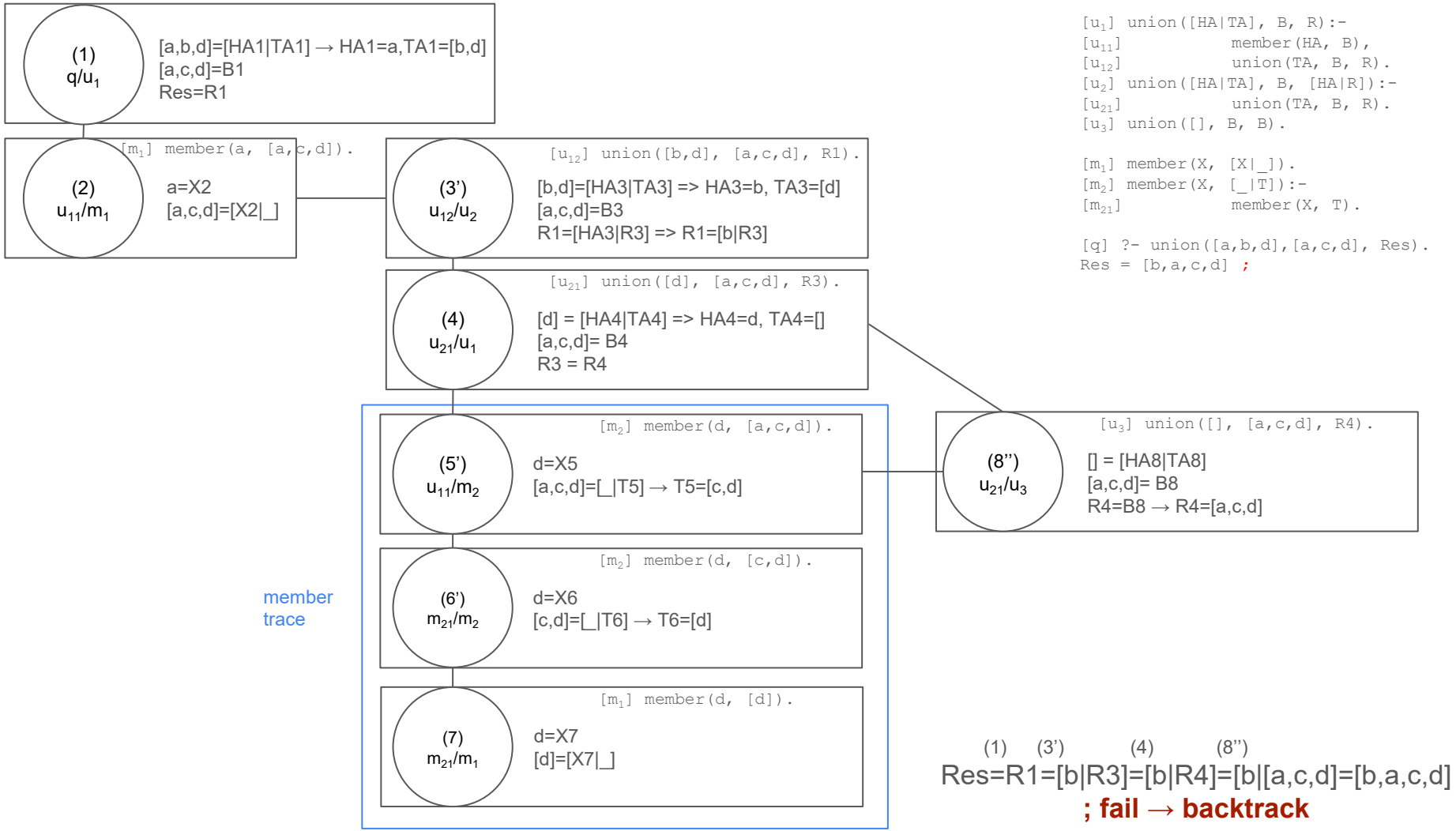

(1)
q/u1
[a,b,d]=[HA1|TA1] → HA1=a,TA1=[b,d]
[a,c,d]=B1
Res=R1
[m1] member(a, [a,c,d]).
(2)
u11/m1
a=X2
[a,c,d]=[X2|_]
[u12] union([b,d], [a,c,d], R1).
(3')
u12/u2
[b,d]=[HA3|TA3] => HA3=b, TA3=[d]
[a,c,d]=B3
R1=[HA3|R3] => R1=[b|R3]
[u21] union([d], [a,c,d], R3).
(4)
u21/u1
[d] = [HA4|TA4] => HA4=d, TA4=[]
[a,c,d]= B4
R3 = R4
[m2] member(d, [a,c,d]).
(5')
u11/m2
d=X5
[a,c,d]=[_|T5] → T5=[c,d]
[m2] member(d, [c,d]).
(6')
m21/m2
d=X6
[c,d]=[_|T6] → T6=[d]
[m1] member(d, [d]).
(7)
m21/m1
d=X7
[d]=[X7|_]
member
trace
[u1] union([], [a,c,d], R4).
(8)
u21/u1
[] = [HA8|TA8]
fail
[a,c,d]= B8
R4=R8
[u1] union([HA|TA], B, R):-
[u11] member(HA, B),
[u12] union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21] union(TA, B, R).
[u3] union([], B, B).
[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21] member(X, T).
[q] ?- union([a,b,d],[a,c,d], Res).

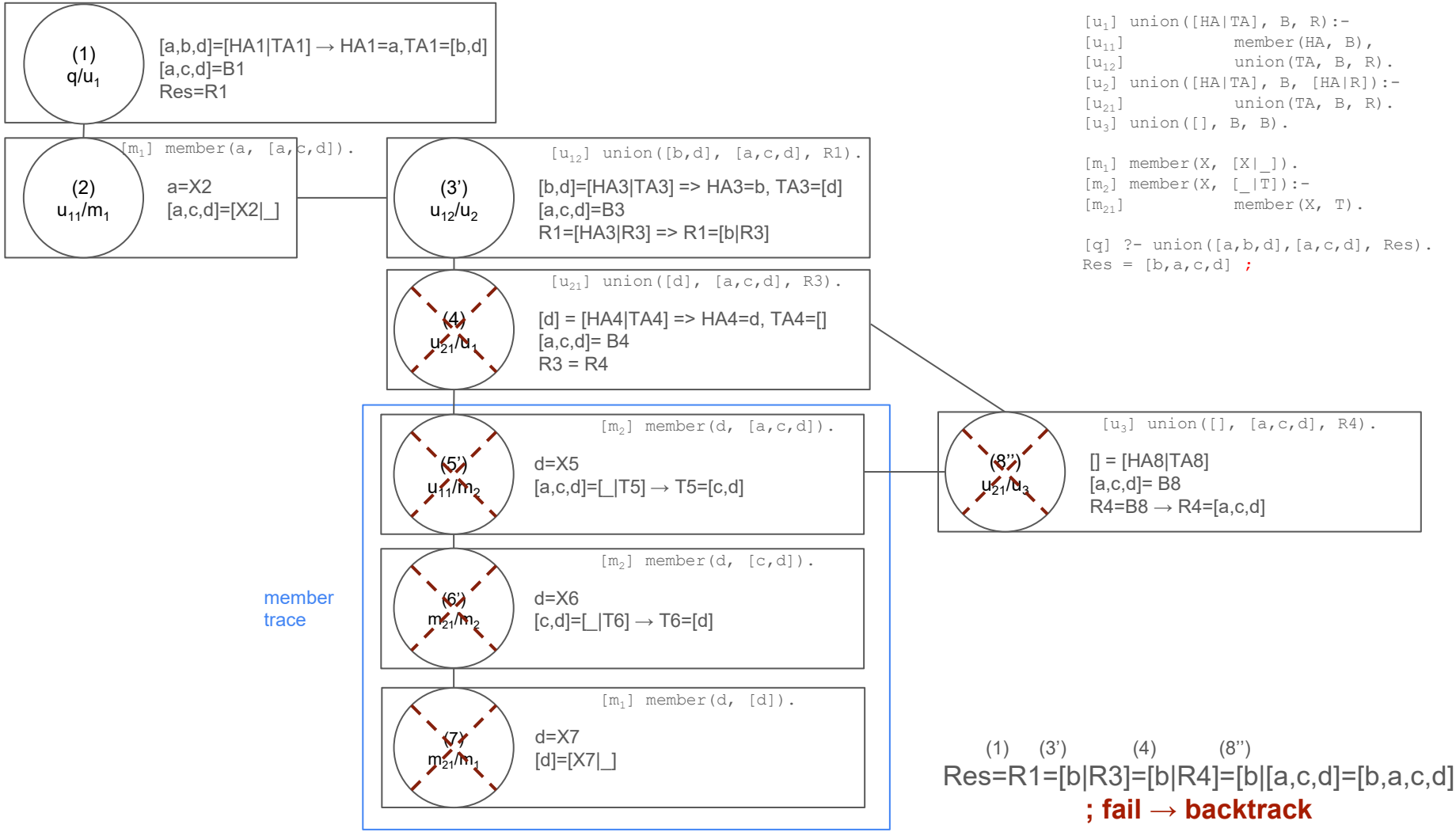

(1)
q/u1
[a,b,d]=[HA1|TA1] → HA1=a,TA1=[b,d]
[a,c,d]=B1
Res=R1
[m1] member(a, [a,c,d]).
(2)
u11/m1
a=X2
[a,c,d]=[X2|_]
[u12] union([b,d], [a,c,d], R1).
(3')
u12/u2
[b,d]=[HA3|TA3] => HA3=b, TA3=[d]
[a,c,d]=B3
R1=[HA3|R3] => R1=[b|R3]
[u21] union([d], [a,c,d], R3).
(4)
u21/u1
[d] = [HA4|TA4] => HA4=d, TA4=[]
[a,c,d]= B4
R3 = R4
[m2] member(d, [a,c,d]).
(5')
u11/m2
d=X5
[a,c,d]=[_|T5] → T5=[c,d]
[m2] member(d, [c,d]).
(6')
m21/m2
d=X6
[c,d]=[_|T6] → T6=[d]
[m1] member(d, [d]).
(7)
m21/m1
d=X7
[d]=[X7|_]
member
trace
[u2] union([], [a,c,d], R4).
(8')
u21/u2
[] = [HA8|TA8]
fail
[a,c,d]= B8
R4=R8
[u1] union([HA|TA], B, R):-
[u11] member(HA, B),
[u12] union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21] union(TA, B, R).
[u3] union([], B, B).
[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21] member(X, T).
[q] ?- union([a,b,d],[a,c,d], Res).

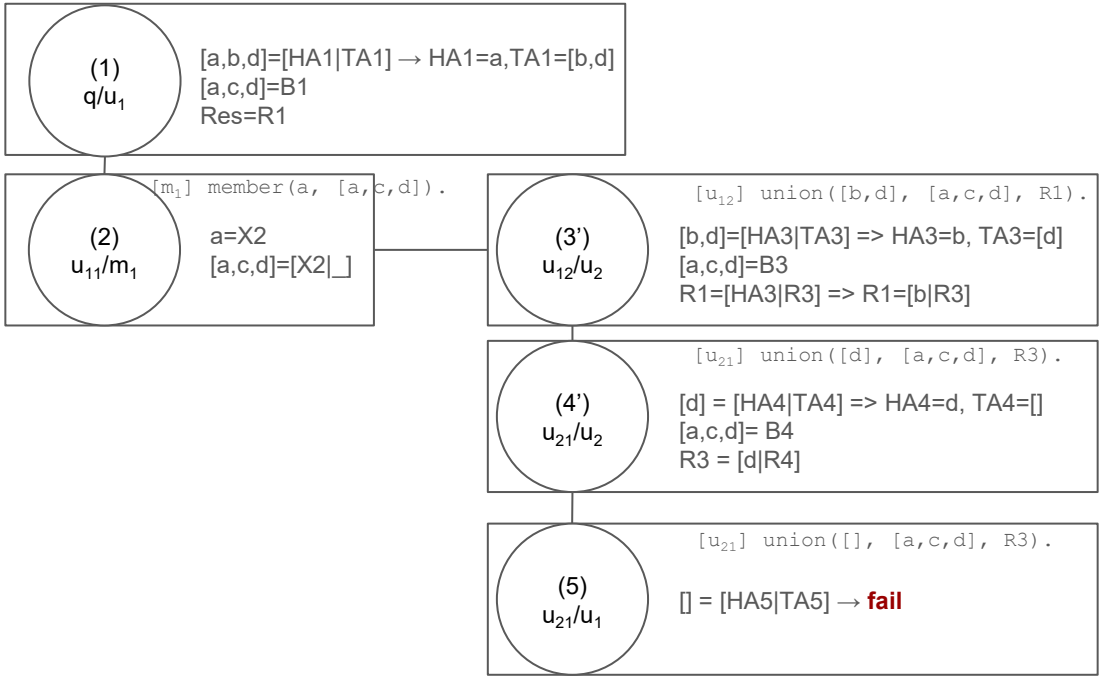

(1)
q/u1
[a,b,d]=[HA1|TA1] → HA1=a,TA1=[b,d]
[a,c,d]=B1
Res=R1
[m1] member(a, [a,c,d]).
(2)
u11/m1
a=X2
[a,c,d]=[X2|_]
[u12] union([b,d], [a,c,d], R1).
(3')
u12/u2
[b,d]=[HA3|TA3] => HA3=b, TA3=[d]
[a,c,d]=B3
R1=[HA3|R3] => R1=[b|R3]
[u21] union([d], [a,c,d], R3).
(4)
u21/u1
[d] = [HA4|TA4] => HA4=d, TA4=[]
[a,c,d]= B4
R3 = R4
[m2] member(d, [a,c,d]).
(5')
u11/m2
d=X5
[a,c,d]=[_|T5] → T5=[c,d]
[m2] member(d, [c,d]).
(6')
m21/m2
d=X6
[c,d]=[_|T6] → T6=[d]
[m1] member(d, [d]).
(7)
m21/m1
d=X7
[d]=[X7|_]
member
trace
[u3] union([], [a,c,d], R4).
(8'')
u21/u3
[] = [HA8|TA8]
[a,c,d]= B8
R4=B8 → R4=[a,c,d]
[u1] union([HA|TA], B, R):-
[u11] member(HA, B),
[u12] union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21] union(TA, B, R).
[u3] union([], B, B).
[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21] member(X, T).
[q] ?- union([a,b,d],[a,c,d], Res).
(1) (3') (4) (8'')
Res=R1=[b|R3]=[b|R4]=[b|[a,c,d]=[b,a,c,d]

Union predicate - backtracking

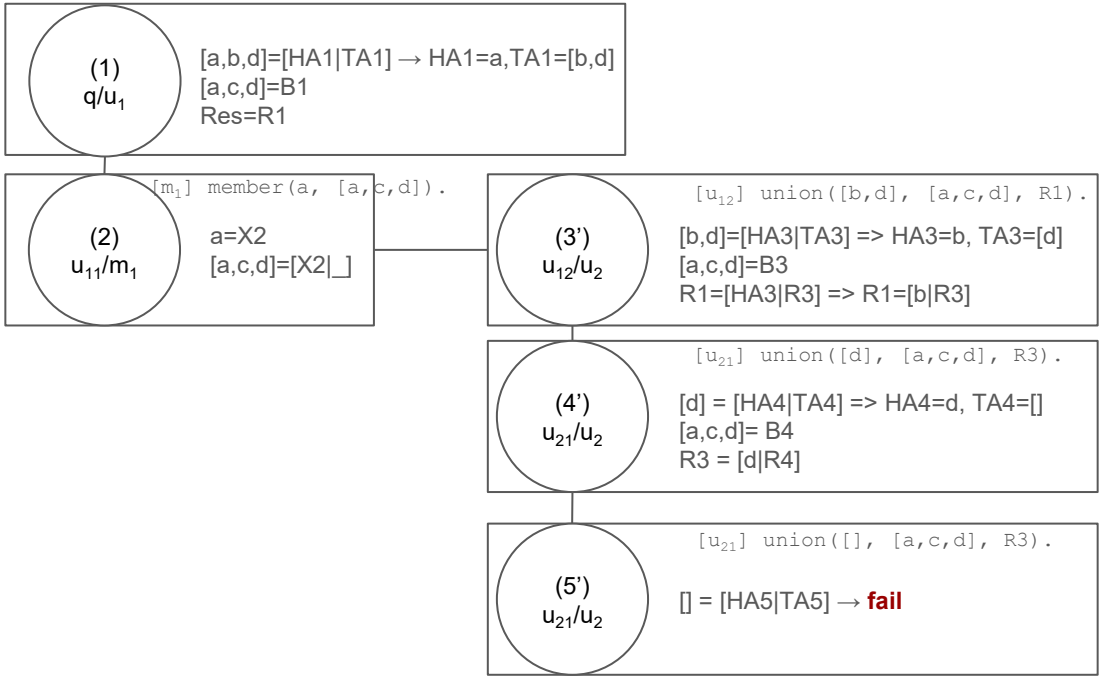
- What happens if we repeat the question?
  - It backtracks and we obtain other solutions (ex. Res=[b,d,a,c,d]) because we do not make the *member/2* check in the alternative clause.
  - When backtracking, it will return first to the *union([d],[a,c,d],R3)* which will merge with u_2 and will create node **4'**.


(1)
q/u1
[a,b,d]=[HA1|TA1] → HA1=a,TA1=[b,d]
[a,c,d]=B1
Res=R1
[m1] member(a, [a,c,d]).
(2)
u11/m1
a=X2
[a,c,d]=[X2|_]
[u12] union([b,d], [a,c,d], R1).
(3')
u12/u2
[b,d]=[HA3|TA3] => HA3=b, TA3=[d]
[a,c,d]=B3
R1=[HA3|R3] => R1=[b|R3]
[u21] union([d], [a,c,d], R3).
(4)
u21/u1
[d] = [HA4|TA4] => HA4=d, TA4=[]
[a,c,d]= B4
R3 = R4
[m2] member(d, [a,c,d]).
(5')
u11/m2
d=X5
[a,c,d]=[_|T5] → T5=[c,d]
[m2] member(d, [c,d]).
(6')
m21/m2
d=X6
[c,d]=[_|T6] → T6=[d]
[m1] member(d, [d]).
(7)
m21/m1
d=X7
[d]=[X7|_]
member
trace
[u3] union([], [a,c,d], R4).
(8'')
u21/u3
[] = [HA8|TA8]
[a,c,d]= B8
R4=B8 → R4=[a,c,d]
[u1] union([HA|TA], B, R):-
[u11] member(HA, B),
[u12] union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21] union(TA, B, R).
[u3] union([], B, B).
[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21] member(X, T).
[q] ?- union([a,b,d],[a,c,d], Res).
Res = [b,a,c,d] ;
(1) (3') (4) (8'')
Res=R1=[b|R3]=[b|R4]=[b|[a,c,d]=[b,a,c,d]
; fail → backtrack

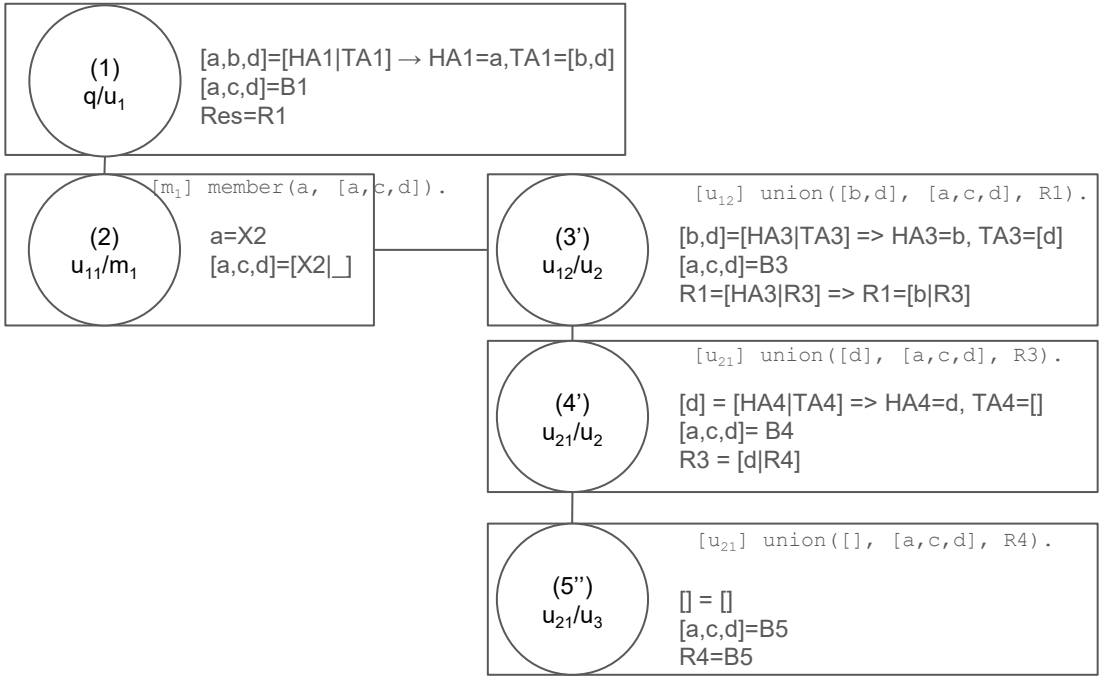

(1)
q/u1
[a,b,d]=[HA1|TA1] → HA1=a,TA1=[b,d]
[a,c,d]=B1
Res=R1
[m1] member(a, [a,c,d]).
(2)
u11/m1
a=X2
[a,c,d]=[X2|_]
[u12] union([b,d], [a,c,d], R1).
(3')
u12/u2
[b,d]=[HA3|TA3] => HA3=b, TA3=[d]
[a,c,d]=B3
R1=[HA3|R3] => R1=[b|R3]
[u21] union([d], [a,c,d], R3).
(4)
u21/u1
[d] = [HA4|TA4] => HA4=d, TA4=[]
[a,c,d]= B4
R3 = R4
[m2] member(d, [a,c,d]).
(5')
u11/m2
d=X5
[a,c,d]=[_|T5] → T5=[c,d]
[m2] member(d, [c,d]).
(6')
m21/m2
d=X6
[c,d]=[_|T6] → T6=[d]
[m1] member(d, [d]).
(7)
m21/m1
d=X7
[d]=[X7|_]
member
trace
[u3] union([], [a,c,d], R4).
(8'')
u21/u3
[] = [HA8|TA8]
[a,c,d]= B8
R4=B8 → R4=[a,c,d]
[u1] union([HA|TA], B, R):-
[u11] member(HA, B),
[u12] union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21] union(TA, B, R).
[u3] union([], B, B).
[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21] member(X, T).
[q] ?- union([a,b,d],[a,c,d], Res).
Res = [b,a,c,d] ;
(1) (3') (4) (8'')
Res=R1=[b|R3]=[b|R4]=[b|[a,c,d]=[b,a,c,d]
; fail → backtrack




```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
Res = [b,a,c,d] ;
```




```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
Res = [b,a,c,d] ;
```




```
[u1] union([HA|TA], B, R):-
[u11]           member(HA, B),
[u12]           union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21]           union(TA, B, R).
[u3] union([], B, B).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- union([a,b,d],[a,c,d], Res).
Res = [b,a,c,d] ;
```

(1) (3') (4') (5'')

Res=R1=[b|R3]=[b|[d|R4]]=[b|[d|[a,c,d]]=[b,d,a,c,d]

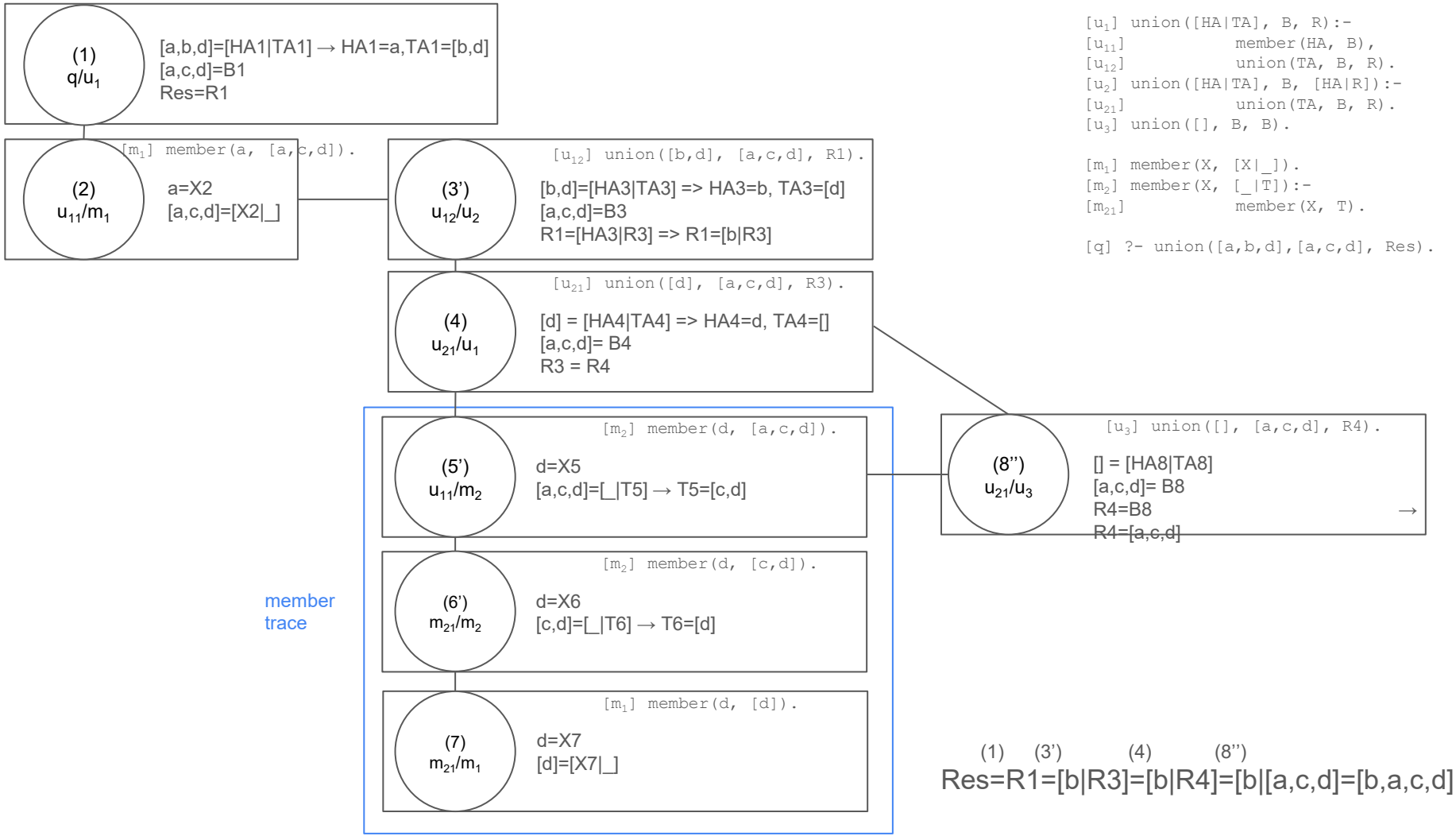
Union predicate - backtracking

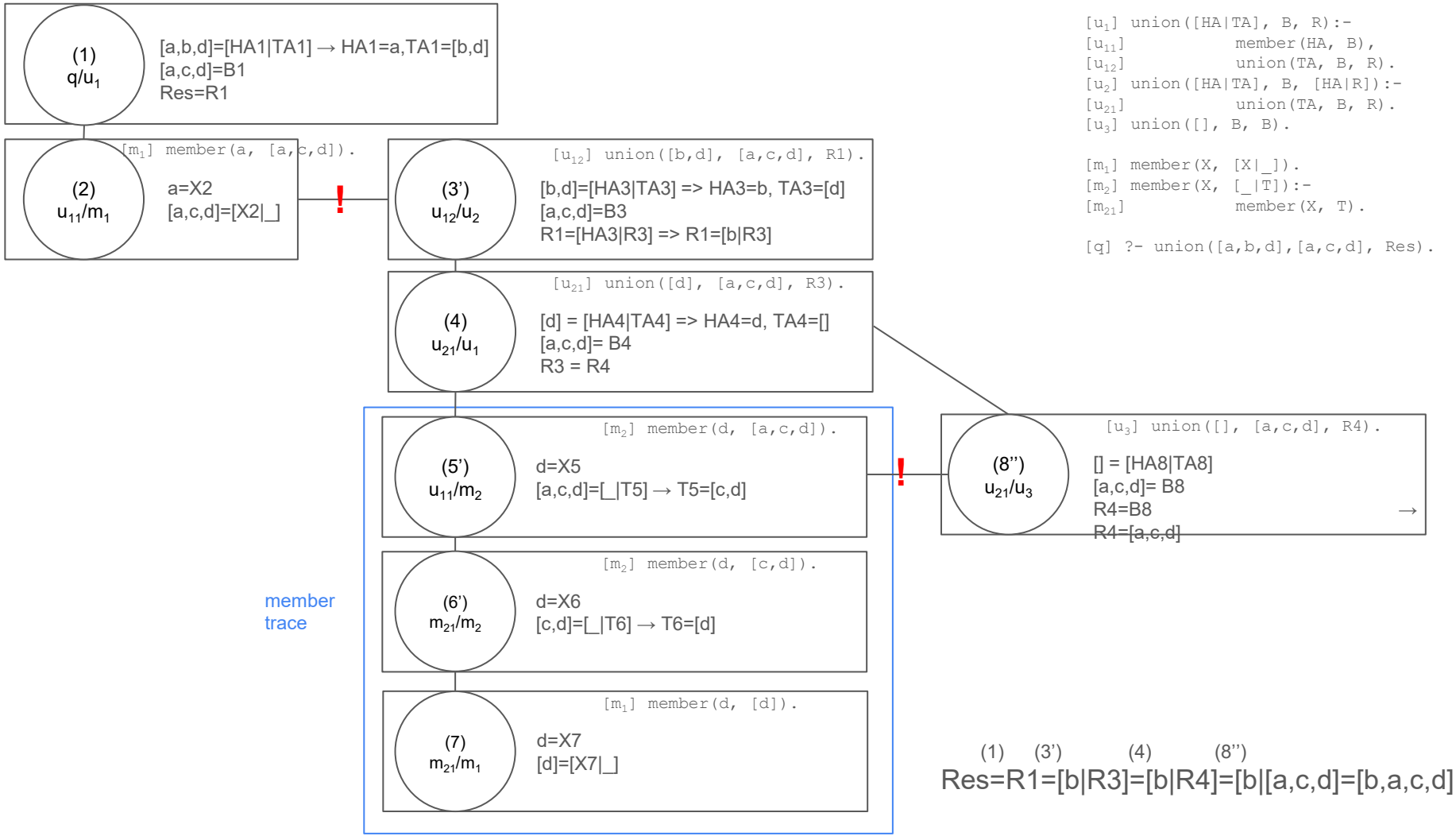
- What happens if we repeat the question?
  - It backtracks and we obtain other solutions (ex. Res=[b,d,a,c,d]) because we do not make the *member/2* check in the alternative clause
  - When backtracking, it will return first to the *union([d],[a,c,d],R3)* which will merge with u_2 and will create node **4'**.
- All answers:
  - Res=[b,a,c,d] ;
  - Res=[b,d,a,c,d] ;
  - Res=[a,b,a,c,d] ;
  - Res=[a,b,d,a,c,d] ;
  - false

Union predicate - the cut

- Adding ! to the predicate

```
[u1]    union([HA|TA], B, R):-
[u11]           member(HA, B), !,
[u12]           union(TA, B, R).
[u2]    union([HA|TA], B, [HA|R]):-
                % not(member(HA, B)),
[u21]           union(TA, B, R).
[u3]    union([], B, B).
```

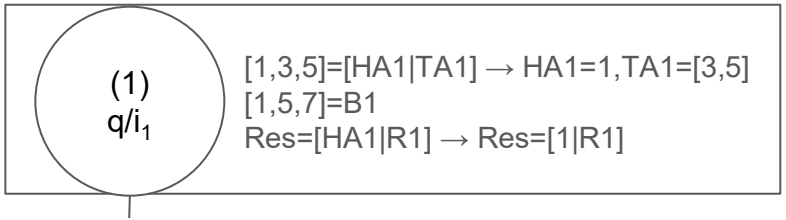

(1)
q/u1
[a,b,d]=[HA1|TA1] → HA1=a,TA1=[b,d]
[a,c,d]=B1
Res=R1
[m1] member(a, [a,c,d]).
(2)
u11/m1
a=X2
[a,c,d]=[X2|_]
[u12] union([b,d], [a,c,d], R1).
(3')
u12/u2
[b,d]=[HA3|TA3] => HA3=b, TA3=[d]
[a,c,d]=B3
R1=[HA3|R3] => R1=[b|R3]
[u21] union([d], [a,c,d], R3).
(4)
u21/u1
[d] = [HA4|TA4] => HA4=d, TA4=[]
[a,c,d]= B4
R3 = R4
[m2] member(d, [a,c,d]).
(5')
u11/m2
d=X5
[a,c,d]=[_|T5] → T5=[c,d]
[m2] member(d, [c,d]).
(6')
m21/m2
d=X6
[c,d]=[_|T6] → T6=[d]
[m1] member(d, [d]).
(7)
m21/m1
d=X7
[d]=[X7|_]
member
trace
[u3] union([], [a,c,d], R4).
(8'')
u21/u3
[] = [HA8|TA8]
[a,c,d]= B8
R4=B8
R4=[a,c,d]
→
[u1] union([HA|TA], B, R):-
[u11] member(HA, B),
[u12] union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21] union(TA, B, R).
[u3] union([], B, B).
[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21] member(X, T).
[q] ?- union([a,b,d],[a,c,d], Res).
(1) (3') (4) (8'')
Res=R1=[b|R3]=[b|R4]=[b|[a,c,d]=[b,a,c,d]


(1)
q/u1
[a,b,d]=[HA1|TA1] → HA1=a,TA1=[b,d]
[a,c,d]=B1
Res=R1
[m1] member(a, [a,c,d]).
(2)
u11/m1
a=X2
[a,c,d]=[X2|_]
!
[u12] union([b,d], [a,c,d], R1).
(3')
u12/u2
[b,d]=[HA3|TA3] => HA3=b, TA3=[d]
[a,c,d]=B3
R1=[HA3|R3] => R1=[b|R3]
[u21] union([d], [a,c,d], R3).
(4)
u21/u1
[d] = [HA4|TA4] => HA4=d, TA4=[]
[a,c,d]= B4
R3 = R4
[m2] member(d, [a,c,d]).
(5')
u11/m2
d=X5
[a,c,d]=[_|T5] → T5=[c,d]
[m2] member(d, [c,d]).
(6')
m21/m2
d=X6
[c,d]=[_|T6] → T6=[d]
[m1] member(d, [d]).
(7)
m21/m1
d=X7
[d]=[X7|_]
member
trace
!
[u3] union([], [a,c,d], R4).
(8'')
u21/u3
[] = [HA8|TA8]
[a,c,d]= B8
R4=B8
R4=[a,c,d]
→
[u1] union([HA|TA], B, R):-
[u11] member(HA, B),
[u12] union(TA, B, R).
[u2] union([HA|TA], B, [HA|R]):-
[u21] union(TA, B, R).
[u3] union([], B, B).
[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21] member(X, T).
[q] ?- union([a,b,d],[a,c,d], Res).
(1) (3') (4) (8'')
Res=R1=[b|R3]=[b|R4]=[b|[a,c,d]=[b,a,c,d]

Intersection predicate

```
[i1]    intersection([HA|TA], B, [HA|R]):-
[i11]           member(HA, B),
[i12]           intersection(TA, B, R).
[i2]    intersection([HA|TA], B, R):-
                % not(member(HA, B)),
[i21]           intersection(TA, B, R).
[i3]    intersection([], _, []).
```

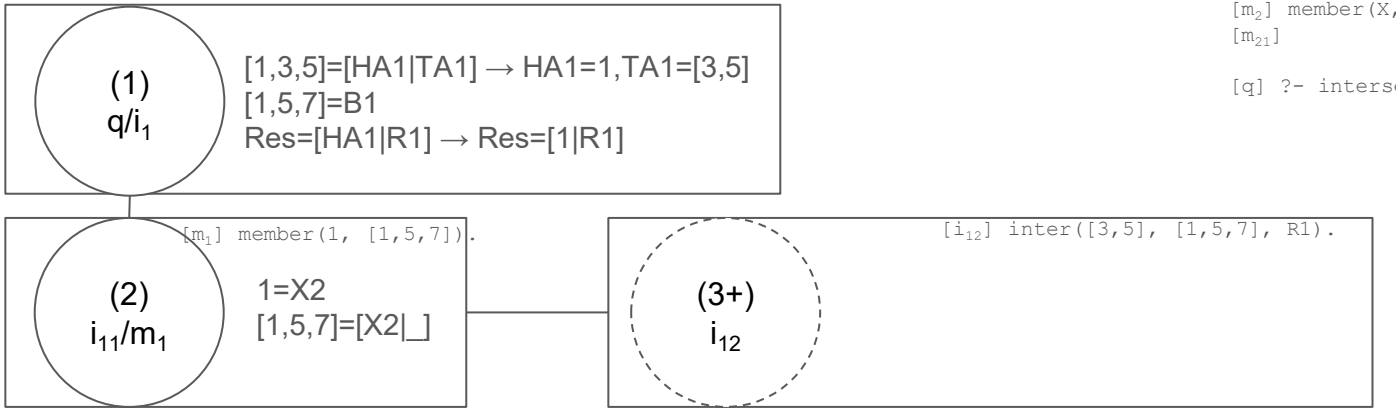
Intersection predicate




```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]           member(HA, B),
[i12]           intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]           intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```

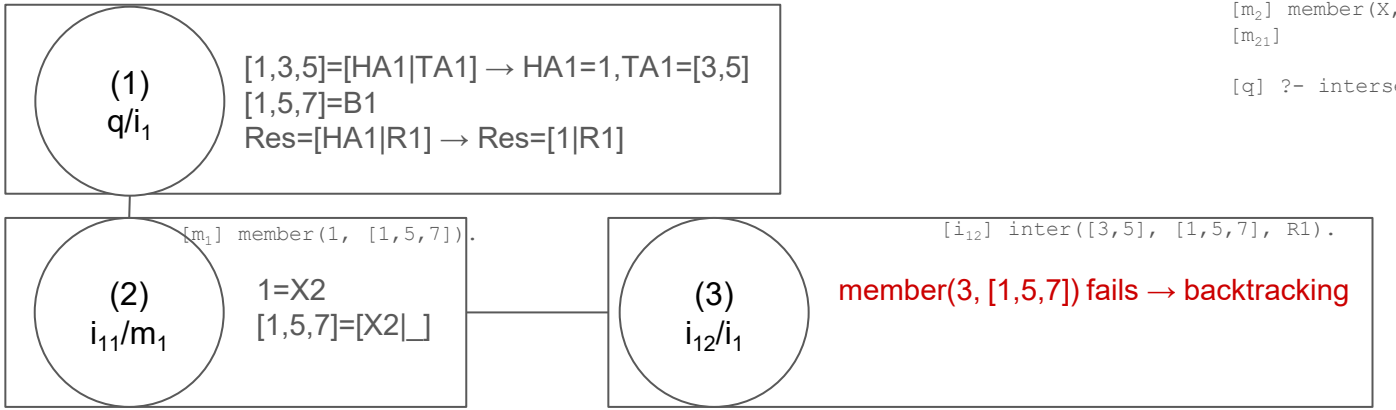
Intersection predicate

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]           member(HA, B),
[i12]           intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]           intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```



Intersection predicate

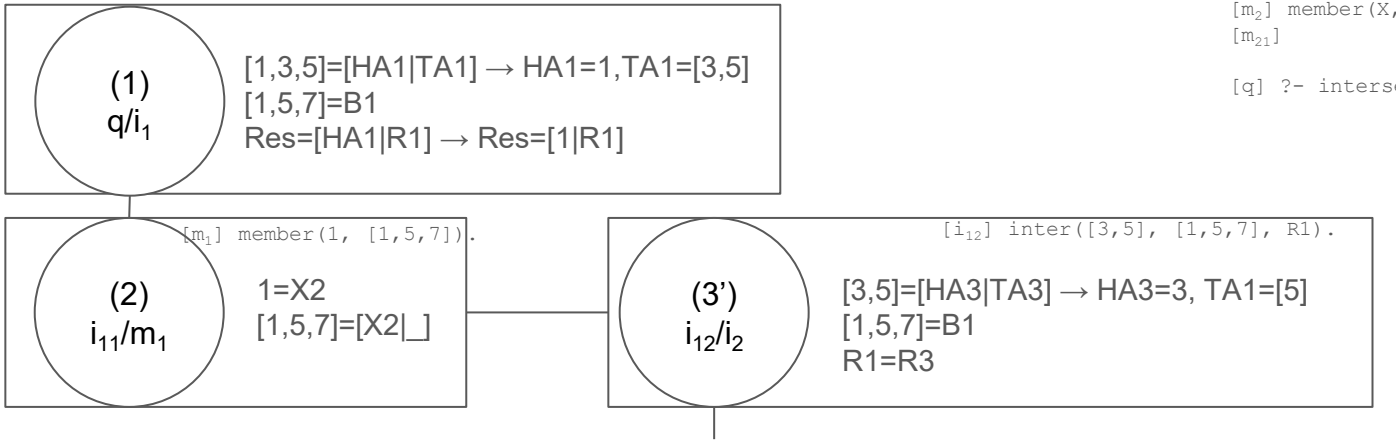
```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]           member(HA, B),
[i12]           intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]           intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```



Suggestion: trace it yourselves and see why and how it fails

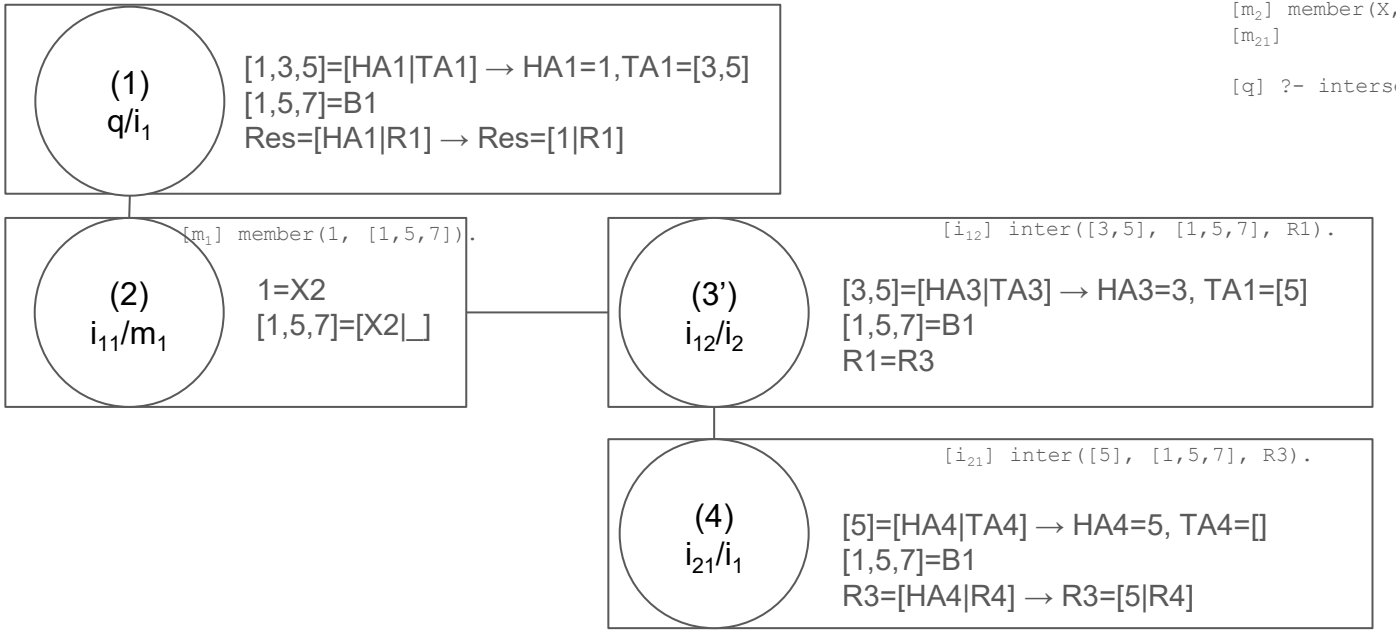
Intersection predicate

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]           member(HA, B),
[i12]           intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]           intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```

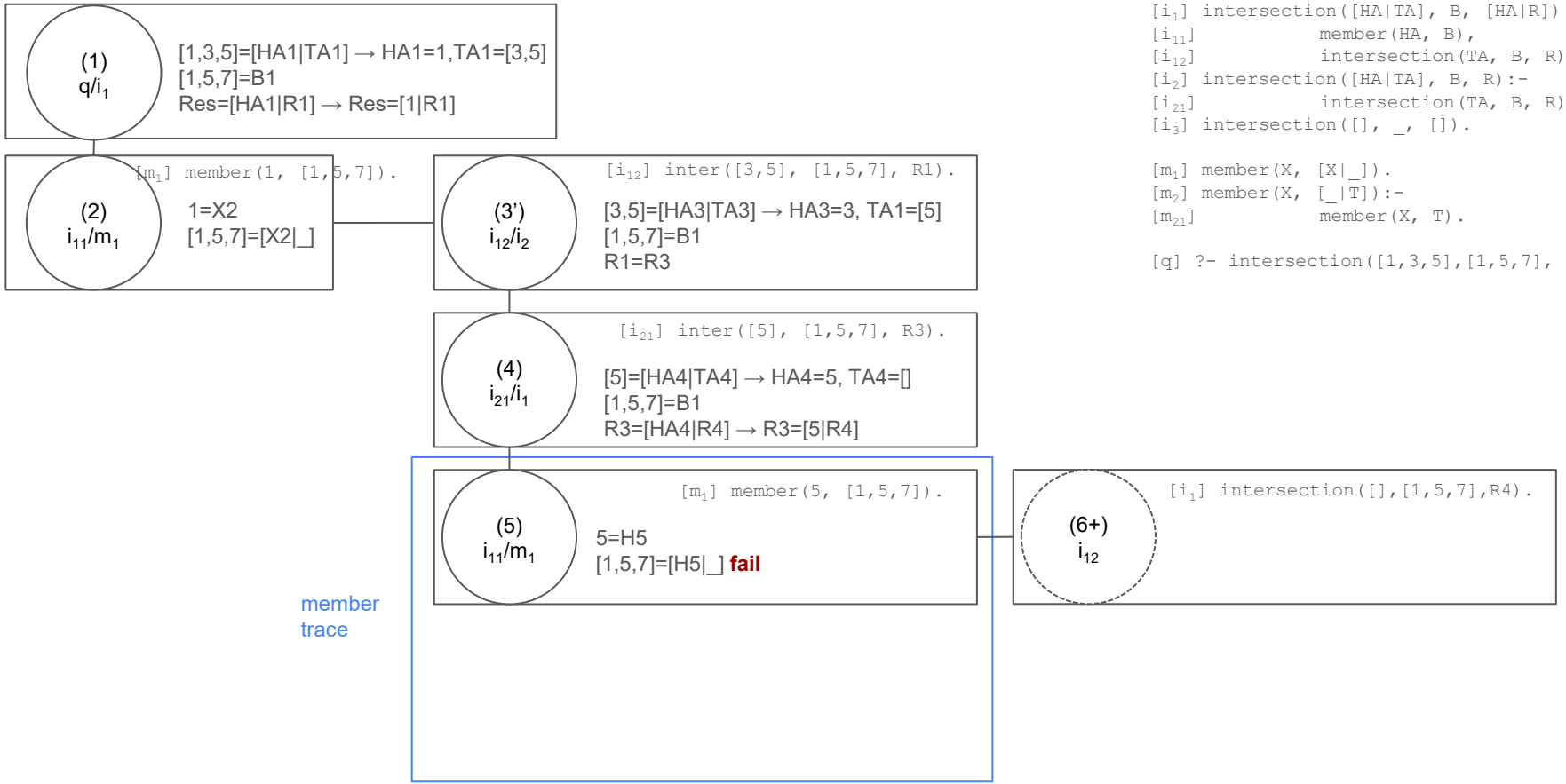


Intersection predicate

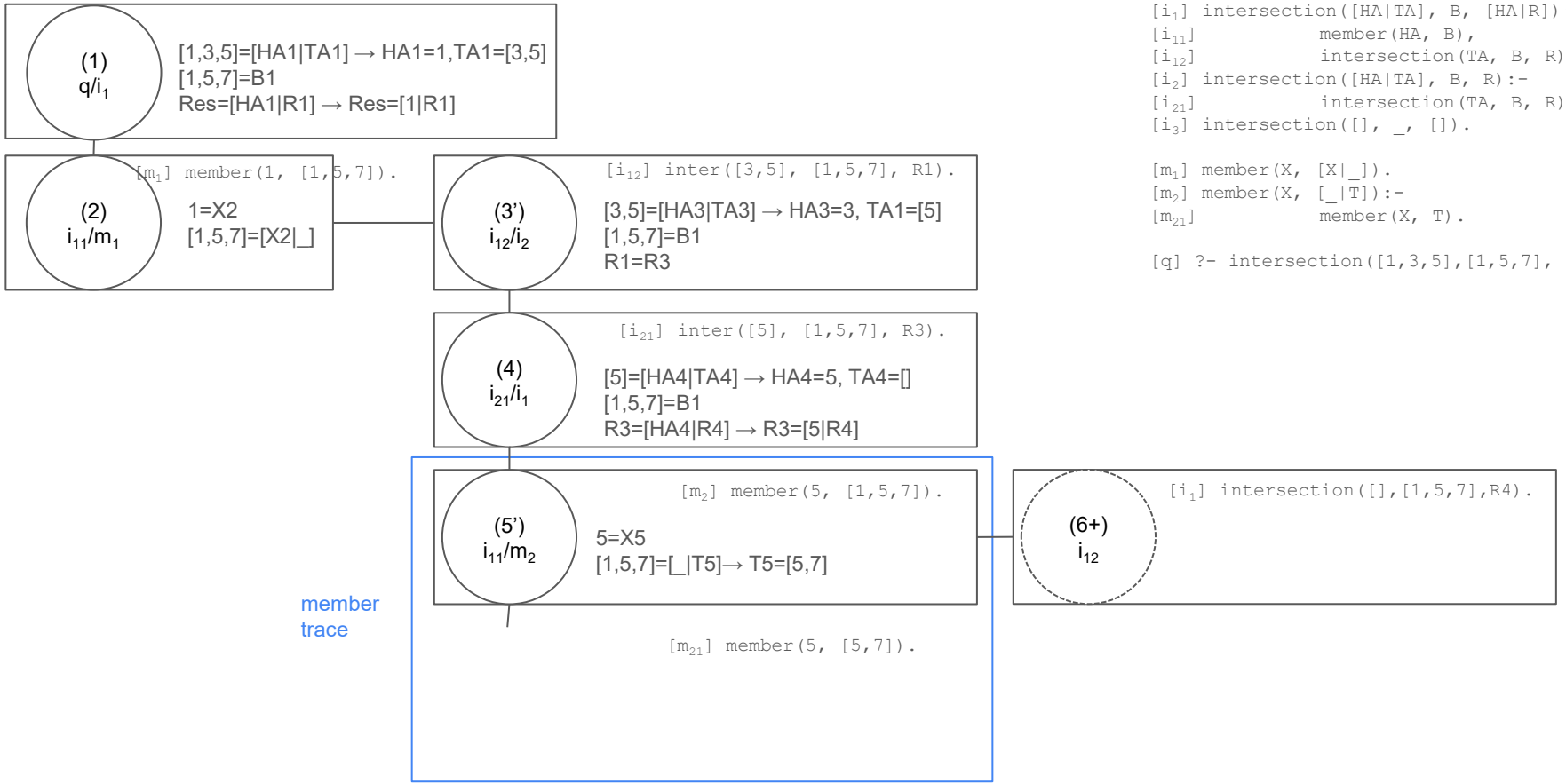
```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]           member(HA, B),
[i12]           intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]           intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```



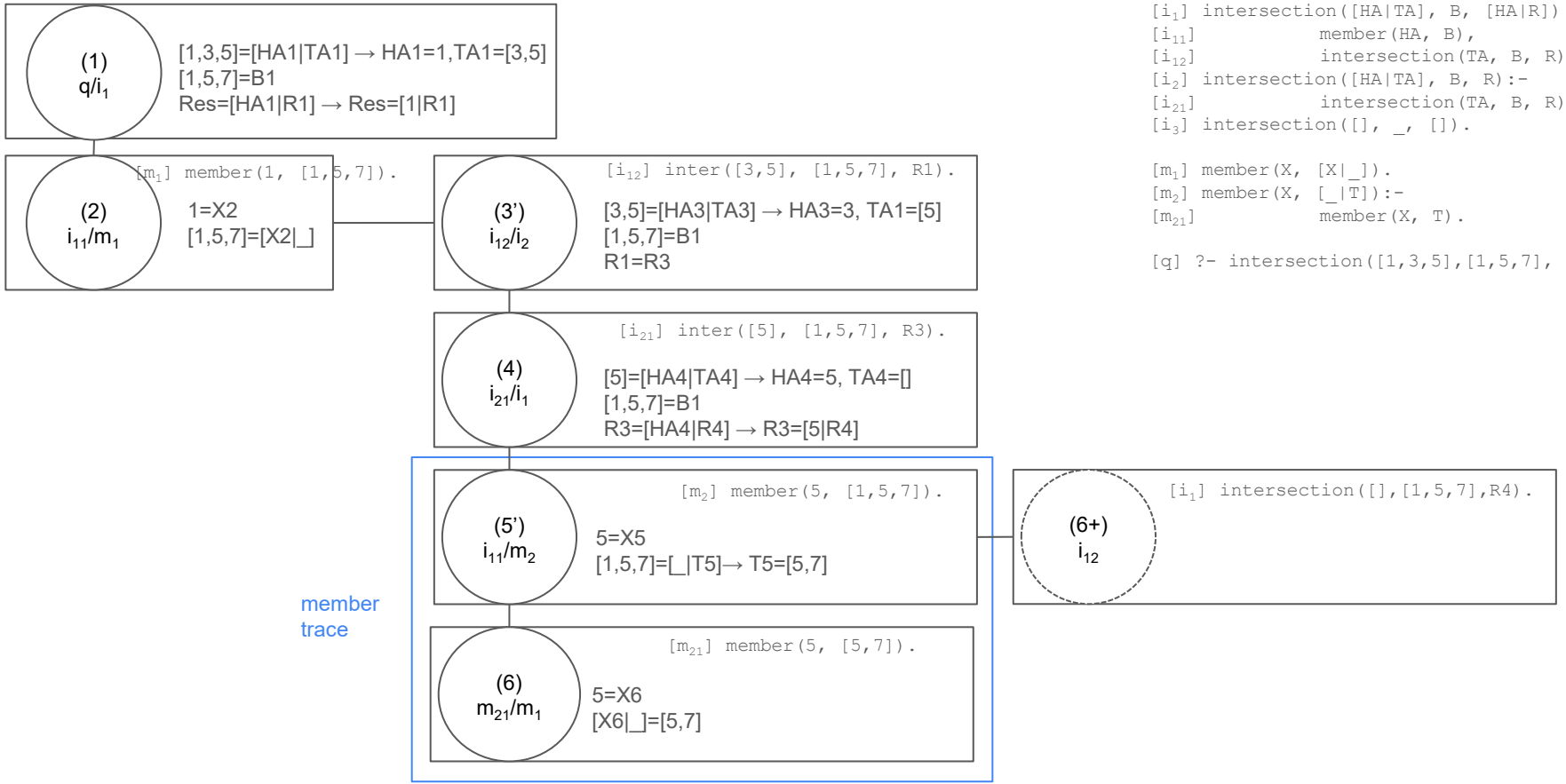

```
[m1] member(5, [1,5,7]).
```



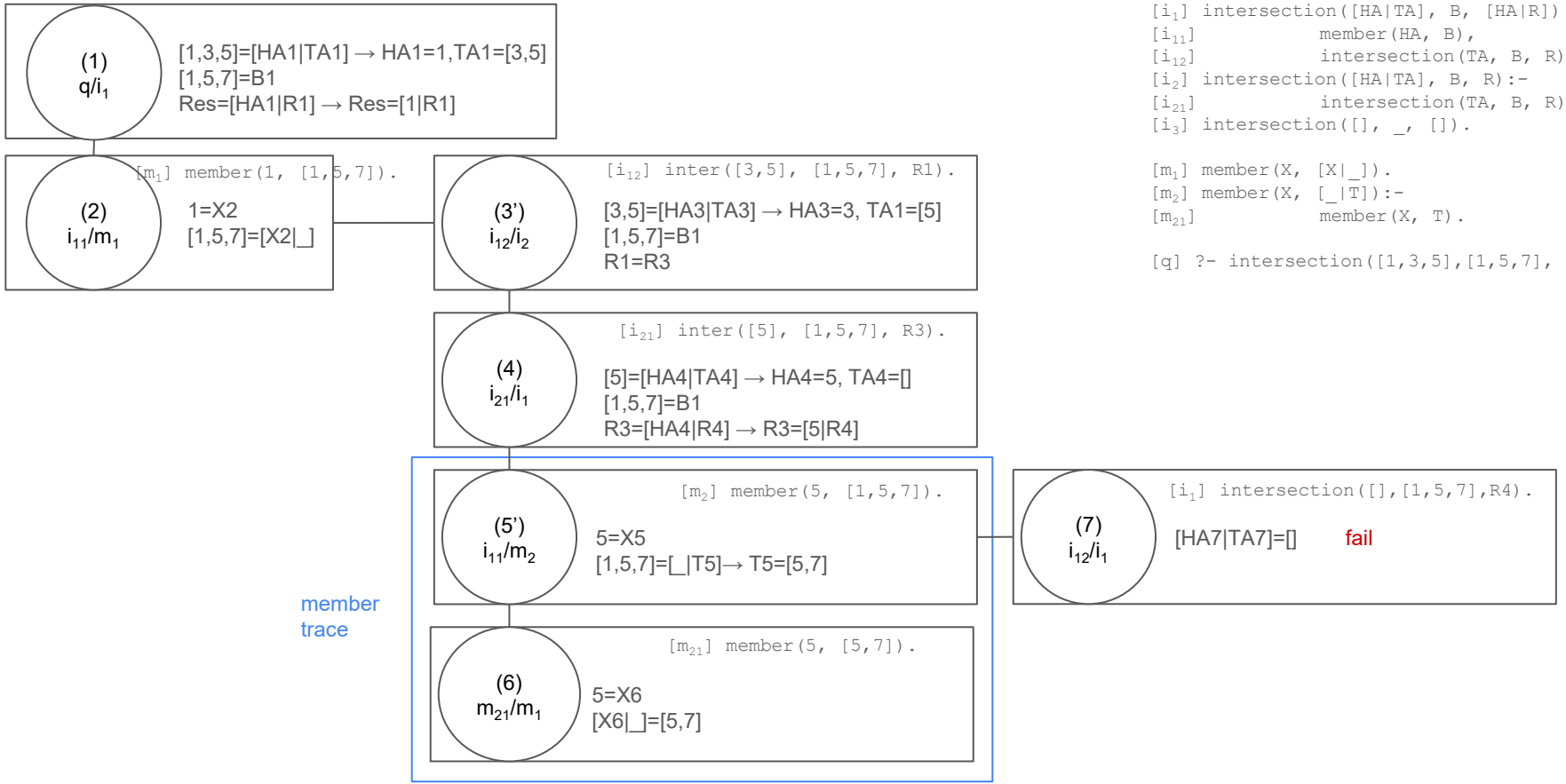

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]             member(HA, B),
[i12]             intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]             intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]             member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```



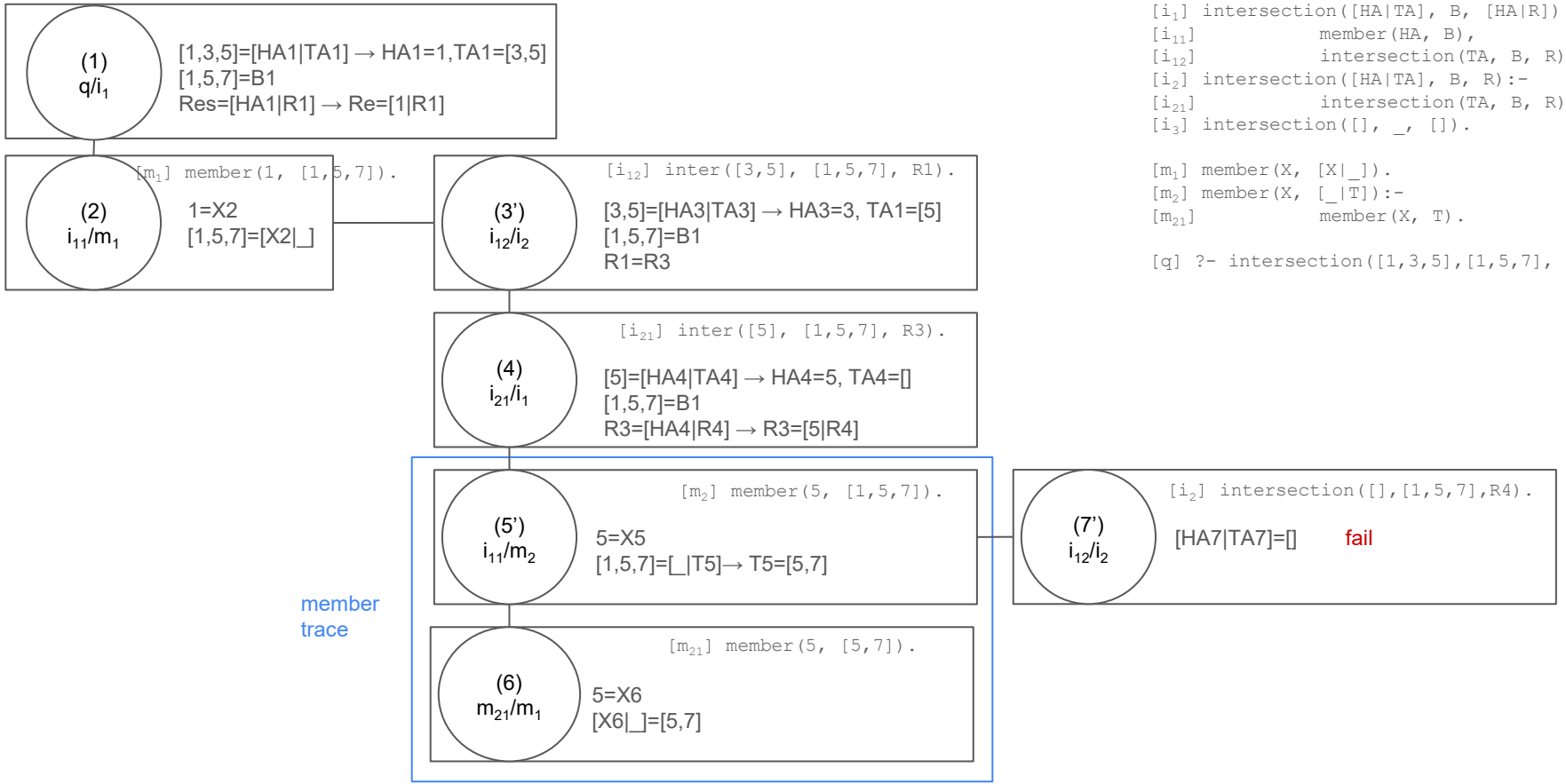

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]              member(HA, B),
[i12]              intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]              intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]              member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```



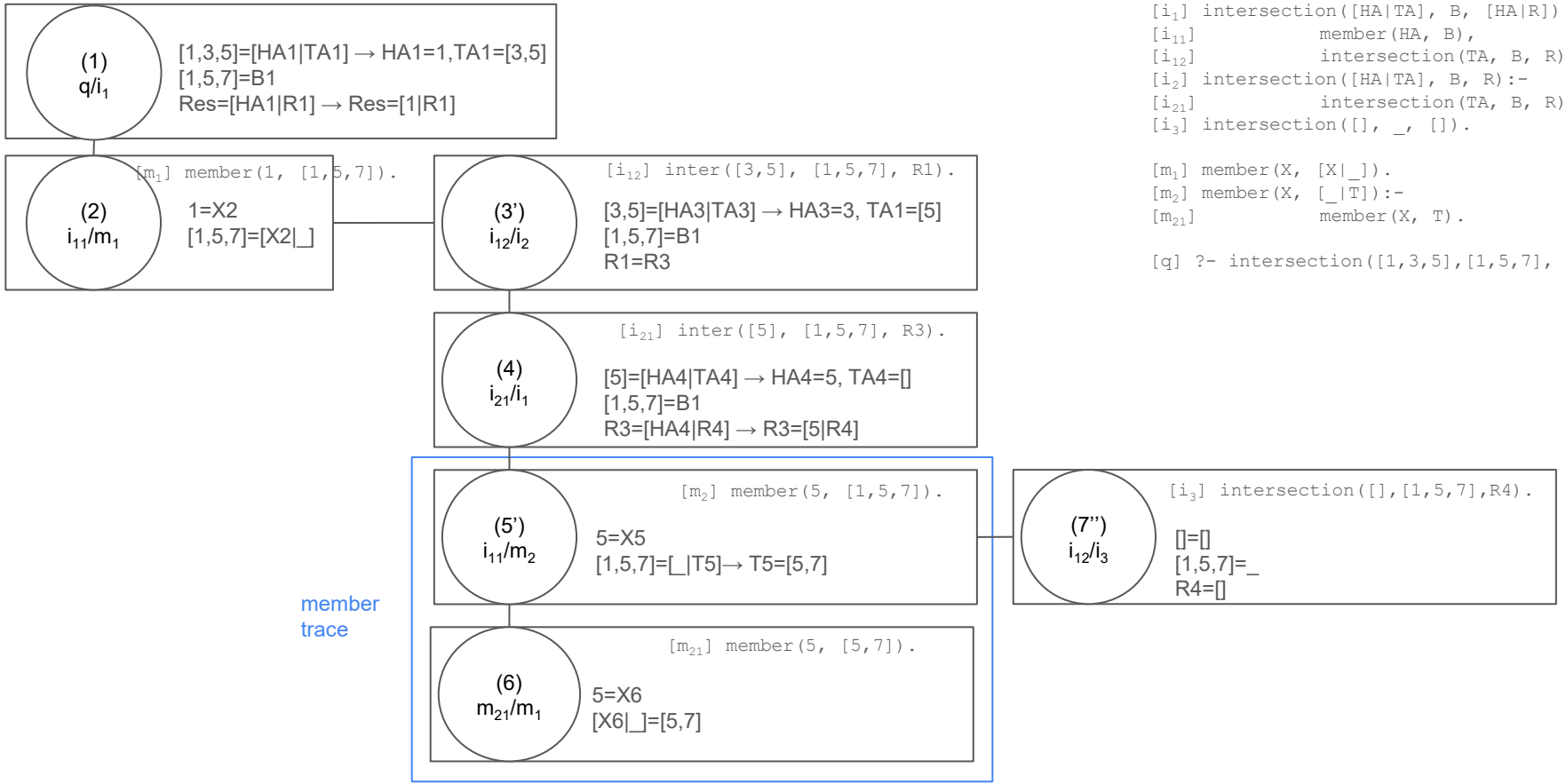

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]           member(HA, B),
[i12]           intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]           intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```



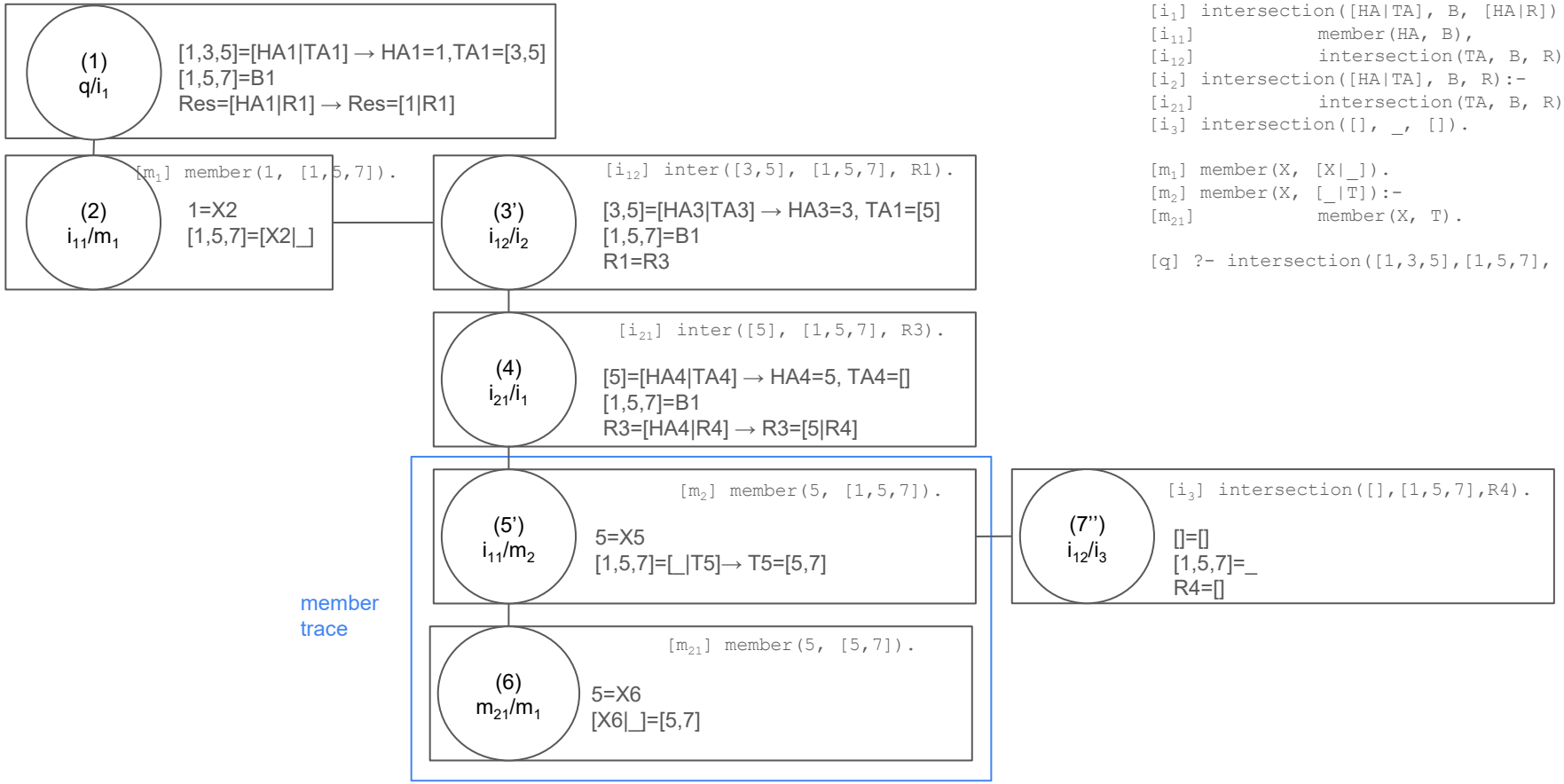

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]             member(HA, B),
[i12]             intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]             intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]             member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```




```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]            member(HA, B),
[i12]            intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]            intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]            member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```



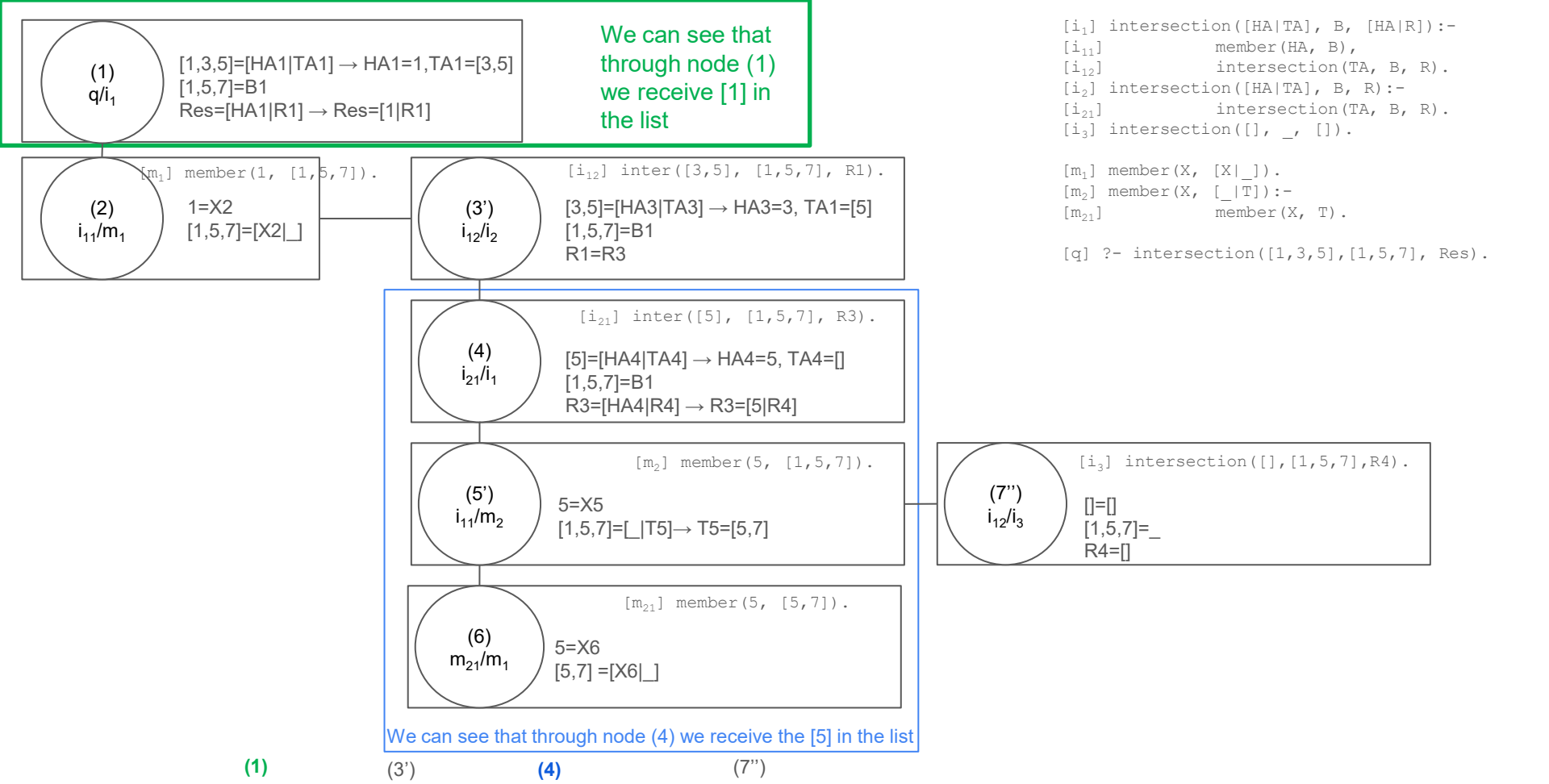

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]            member(HA, B),
[i12]            intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]            intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]            member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```




```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]            member(HA, B),
[i12]            intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]            intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]            member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```

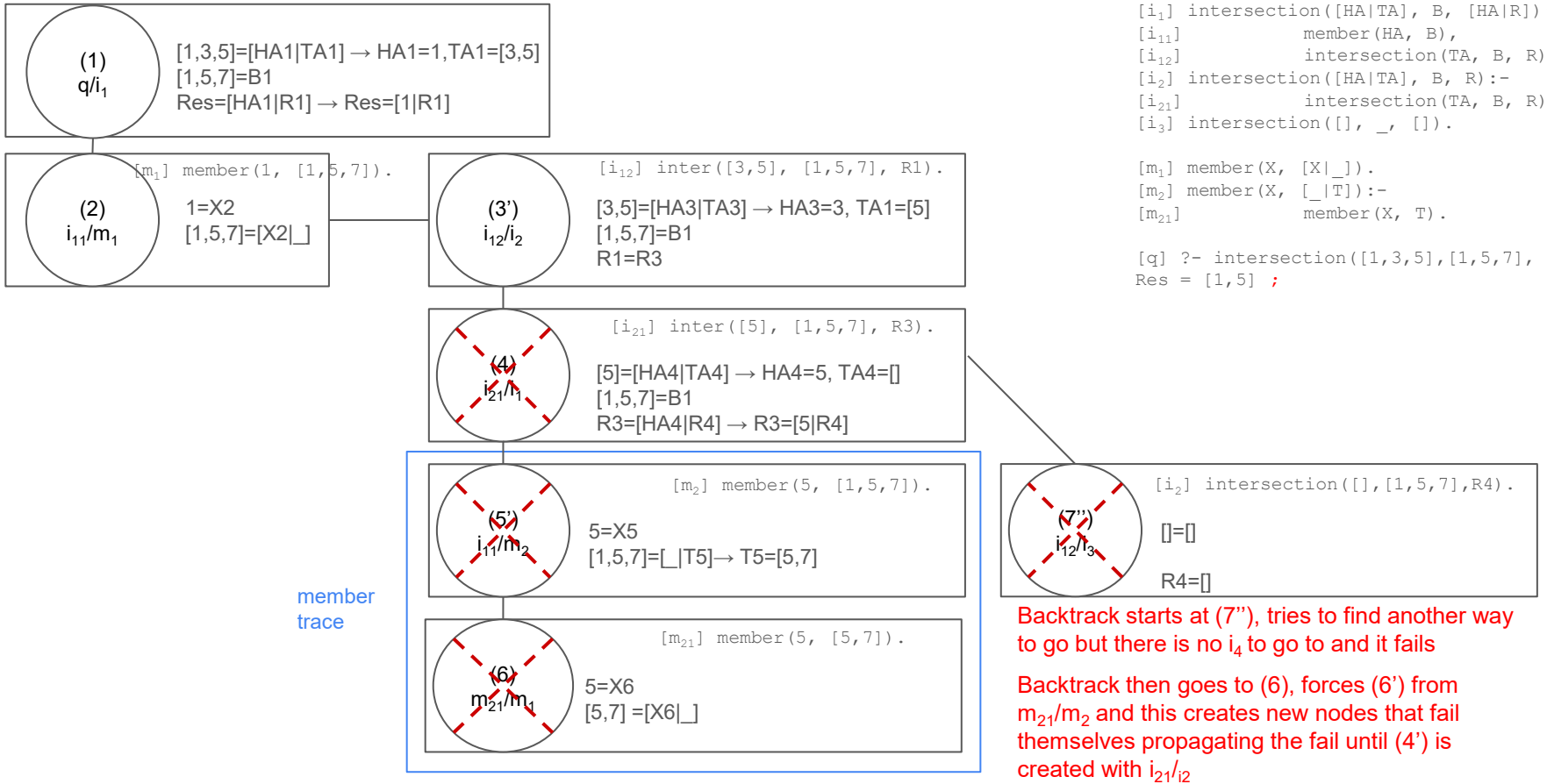
(1) (3') (4) (7'')

Res = [1|R1] = [1|R3] = [1|[5|R4]] = [1|[5|[]]] = [1,5]


(1)
q/i1
[1,3,5]=[HA1|TA1] → HA1=1,TA1=[3,5]
[1,5,7]=B1
Res=[HA1|R1] → Res=[1|R1]
We can see that through node (1) we receive [1] in the list
[i1] intersection([HA|TA], B, [HA|R]):-
[i11] member(HA, B),
[i12] intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21] intersection(TA, B, R).
[i3] intersection([], _, []).
[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21] member(X, T).
[q] ?- intersection([1,3,5],[1,5,7], Res).
[m1] member(1, [1,5,7]).
(2)
i11/m1
1=X2
[1,5,7]=[X2|_]
[i12] inter([3,5], [1,5,7], R1).
(3')
i12/i2
[3,5]=[HA3|TA3] → HA3=3, TA1=[5]
[1,5,7]=B1
R1=R3
[i21] inter([5], [1,5,7], R3).
(4)
i21/i1
[5]=[HA4|TA4] → HA4=5, TA4=[]
[1,5,7]=B1
R3=[HA4|R4] → R3=[5|R4]
[m2] member(5, [1,5,7]).
(5')
i11/m2
5=X5
[1,5,7]=[_|T5]→ T5=[5,7]
[i3] intersection([],[1,5,7],R4).
(7'')
i12/i3
[]=[]
[1,5,7]=_
R4=[]
[m21] member(5, [5,7]).
(6)
m21/m1
5=X6
[5,7] =[X6|_]
We can see that through node (4) we receive the [5] in the list
(1)
(3')
(4)
(7'')
Res = [1|R1] = [1|R3] = [1|[5|R4]] = [1|[5|[]]] = [1,5]

Intersection predicate - backtracking

- Upon repeating the question, we reach through backtracking node 4 which unified with i_1
  - Now it unifies with i_2




```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]              member(HA, B),
[i12]              intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]              intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]              member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
Res = [1,5] ;
```

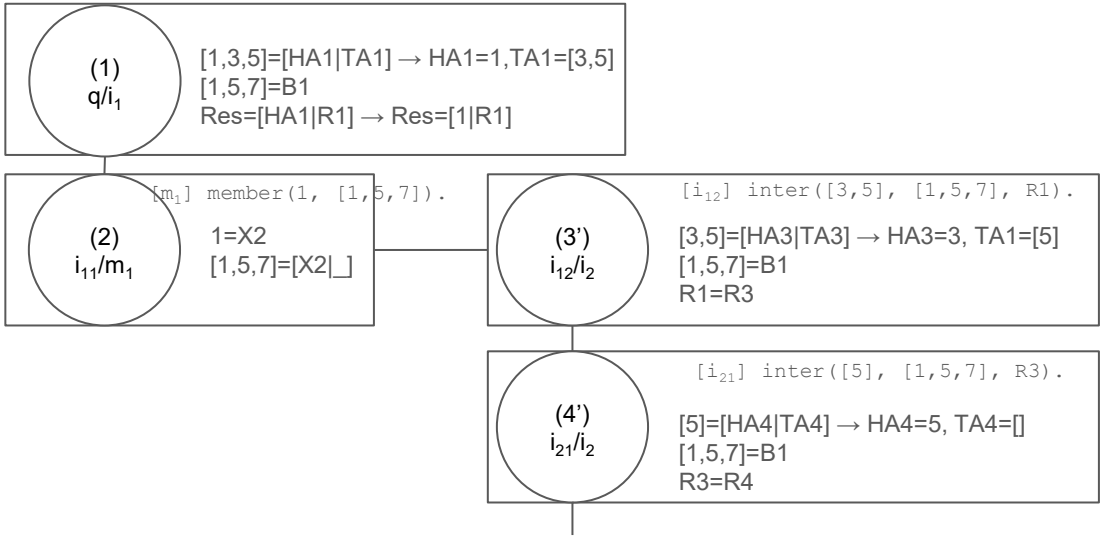
Backtrack starts at (7''), tries to find another way to go but there is no i_4 to go to and it fails

Backtrack then goes to (6), forces (6') from m_{21}/m_2 and this creates new nodes that fail themselves propagating the fail until (4') is created with i_{21}/i_2

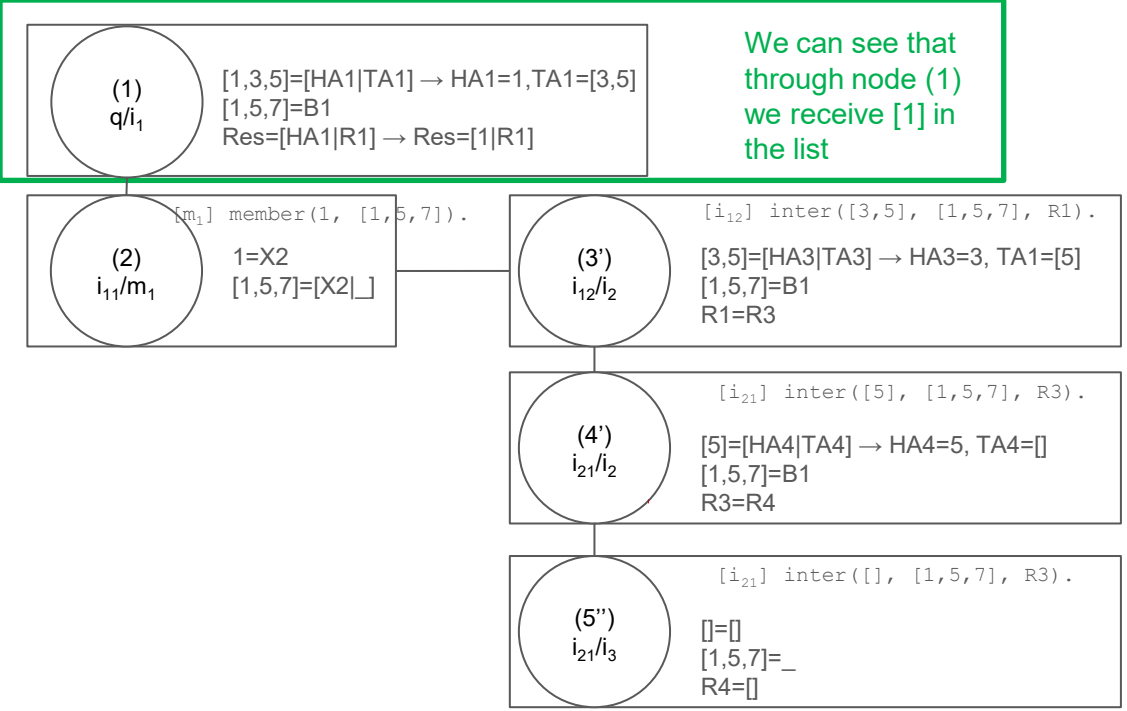
(1) (3') (4) (7'')

Res = [1|R1] = [1|R3] = [1|[5|R4]] = [1|[5|[]]] = [1,5]

; Next → backtrack



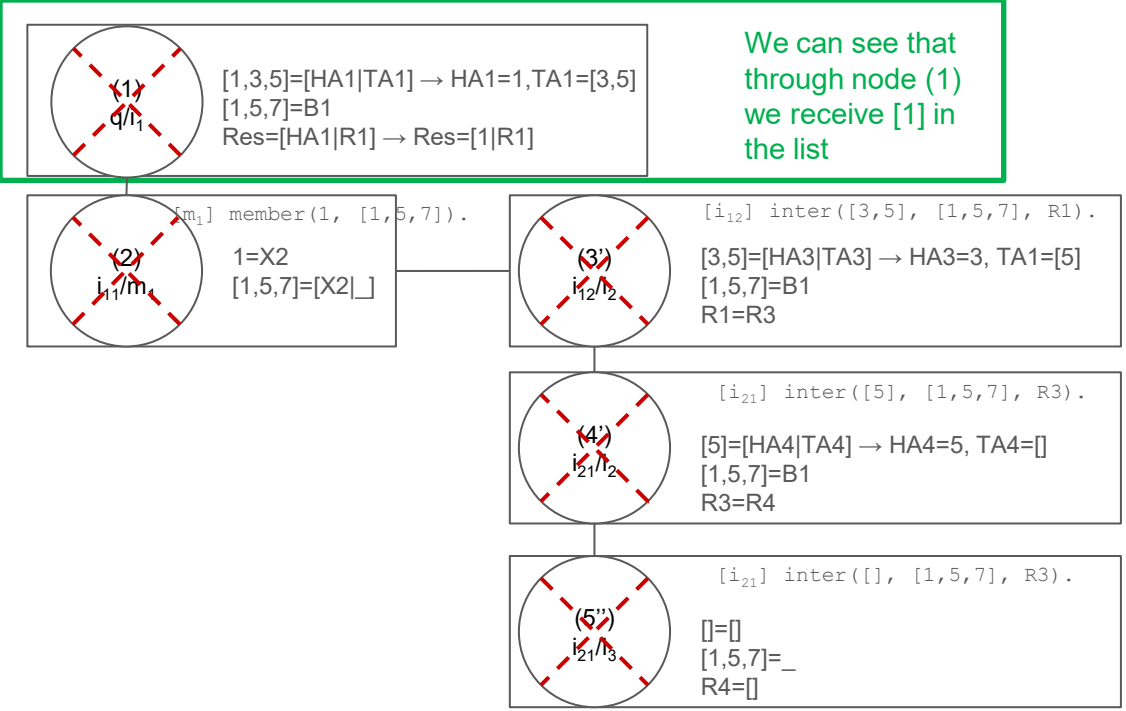

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]           member(HA, B),
[i12]           intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]           intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]           member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
Res = [1,5] ;
```




```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]            member(HA, B),
[i12]            intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]            intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]            member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
Res = [1,5] ;
Res = [1]
```

However, node (4) is lost to backtracking (in favor to (4')), and as such, the [5] is not a part of the list

$$\text{Res} = \overset{(1)}{[1|R1]} = \overset{(3')}{[1|R3]} = \overset{(4')}{[1|R4]} = \overset{(5'')}{[1|[]]} = [1]$$



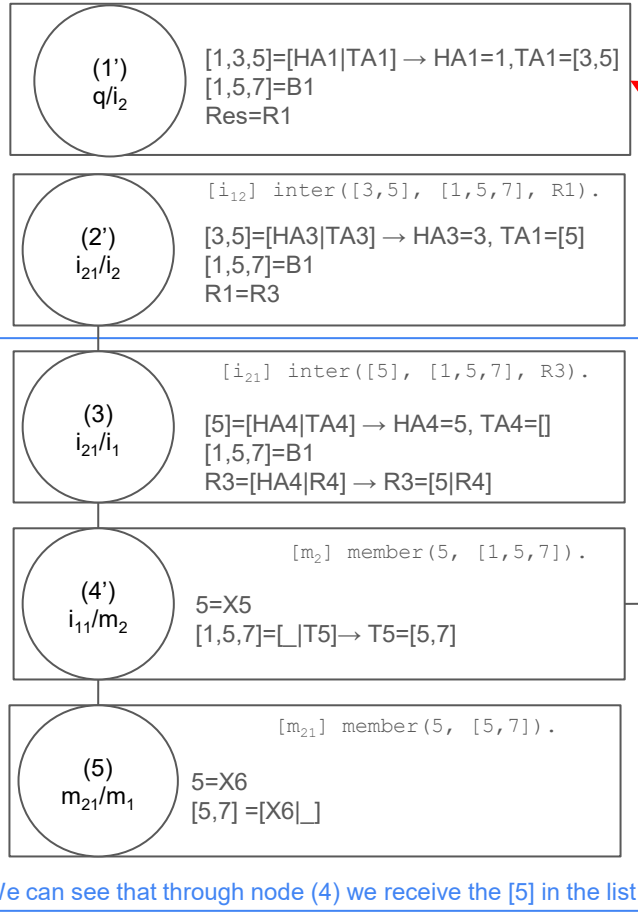

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]             member(HA, B),
[i12]             intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]             intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]             member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
Res = [1,5] ;
Res = [1]
```

Backtrack starts at (5''), tries to find another way to go but there is no i_4 to go to and it fails

Backtrack then goes to (4'), forces (4'') but it fails propagating the fail upwards until (2). The backtrack of node (2) forces (2') from i_{11}/m_2 and this creates new nodes that fail themselves propagating the fail until (1') is created with $q/_{i2}$

$$\text{Res} \overset{(1)}{=} [1|R1] \overset{(3')}{=} [1|R3] \overset{(4')}{=} [1|R4] = [1|[]] \overset{(5'')}{=} [1]$$

; Next → backtrack



We can see that through node (4) we receive the [5] in the list

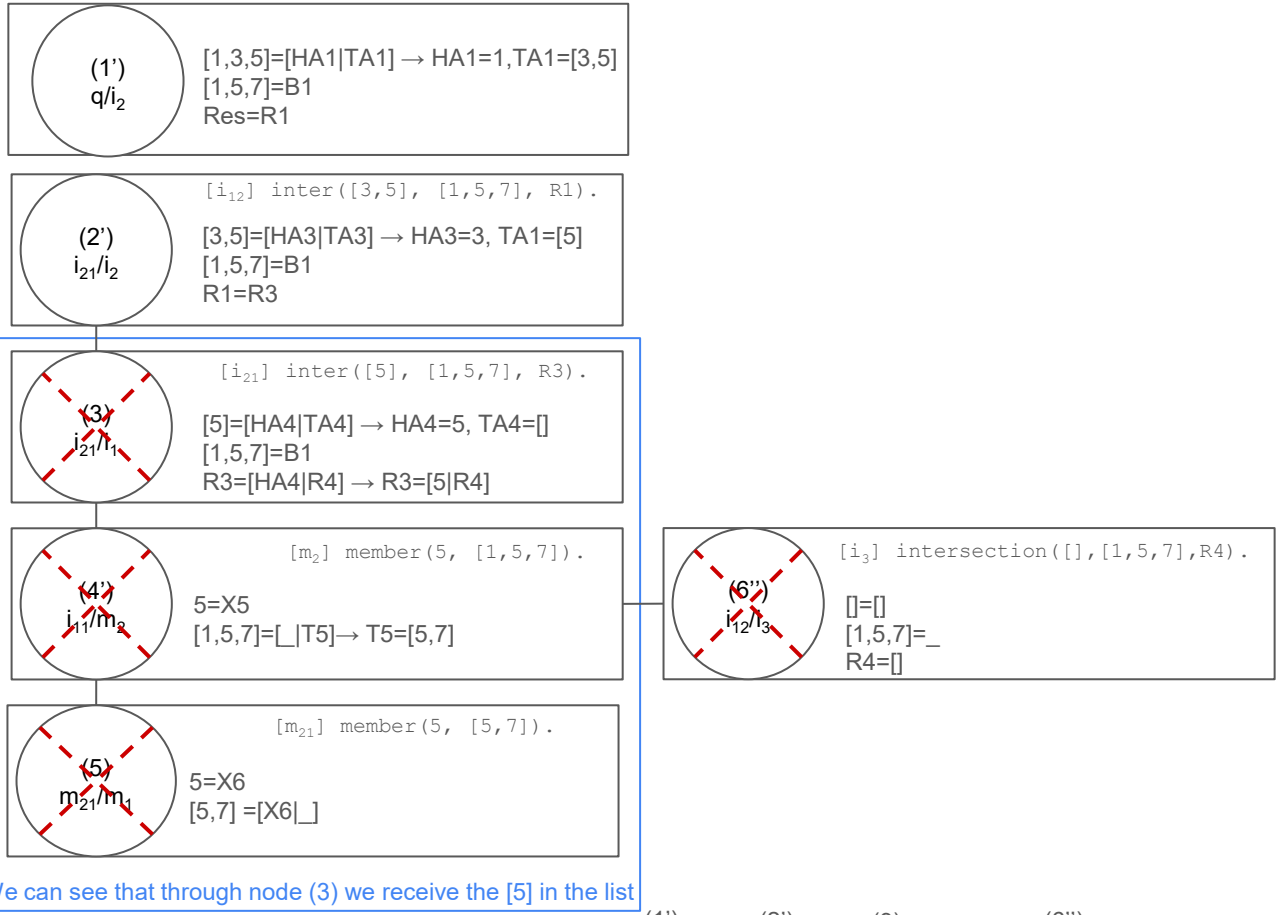
However, node (1) was lost to backtracking (in favor to (1')), and as such, the [1] is not a part of the list

Do notice that because the starting point is new (1') the rest of the tree is the same as the first answer

(1') (2') (3) (6'')

Res = R1 = R3= [5|R4] = [5|[]] = [5]

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]            member(HA, B),
[i12]            intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]            intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]            member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```



We can see that through node (3) we receive the [5] in the list

```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]            member(HA, B),
[i12]            intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]            intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]            member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```

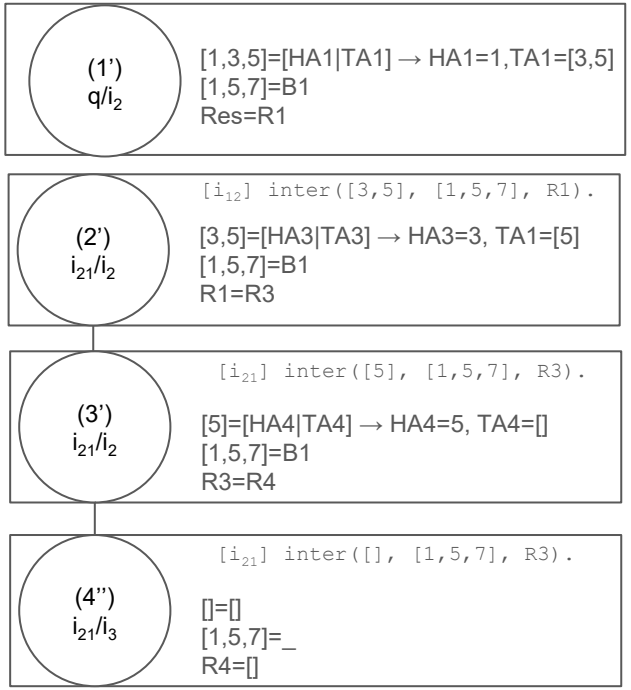
Backtrack starts at (6''), tries to find another way to go but there is no i_4 to go to and it fails

Backtrack then goes to (5), forces (5') from m_{21}/m_2 and this creates new nodes that fail themselves propagating the fail until (3') is created with i_{21}/i_{i2}

(1') (2') (3) (6'')

$$Res = R1 = R3 = [5|R4] = [5|[]] = [5]$$

; Next → backtrack

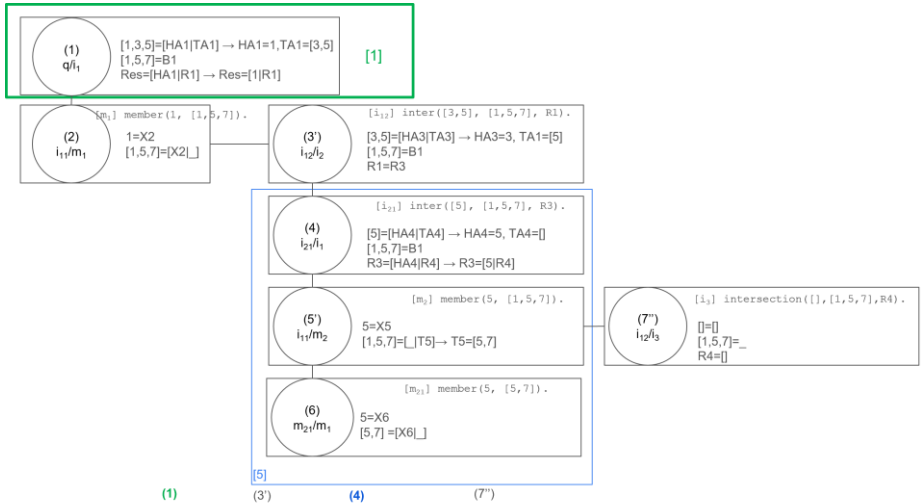



```
[i1] intersection([HA|TA], B, [HA|R]):-
[i11]             member(HA, B),
[i12]             intersection(TA, B, R).
[i2] intersection([HA|TA], B, R):-
[i21]             intersection(TA, B, R).
[i3] intersection([], _, []).

[m1] member(X, [X|_]).
[m2] member(X, [_|T]):-
[m21]             member(X, T).

[q] ?- intersection([1,3,5],[1,5,7], Res).
```

However, node (3) is lost to backtracking (in favor to (3')), and as such, the [5] is not a part of the list

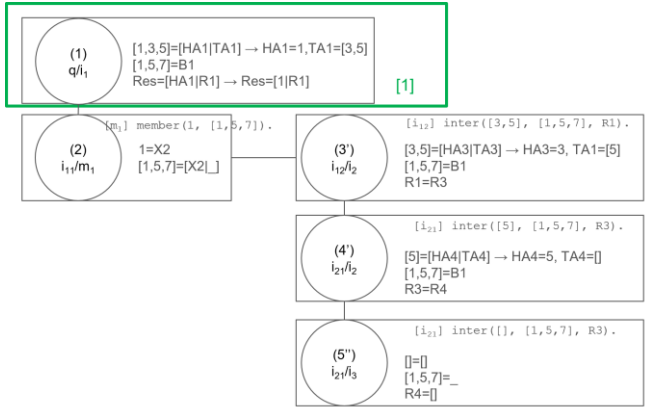
$$\text{Res} \overset{(1')}{=} \text{R1} \overset{(2')}{=} \text{R3} \overset{(3)}{=} \text{R4} \overset{(4'')}{=} []$$

Last ; Next → backtrack, fails from (4'') and propagates until everything is unmade and the last answer is : false.



(1) (3') (4) (7'')

Res = [1|R1] = [1|R3] = [1|[5|R4]] = [1|[5|[]]] = [1,5]

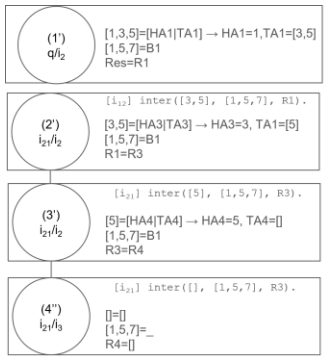


(1) (3') (4') (5'')

Res = [1|R1] = [1|R3] = [1|R4] = [1|[]] = [1]

(1') (2') (3) (6'')

Res = R1 = R3= [5|R4] = [5|[]] = [5]



(1') (2') (3) (4'')

Res = R1 = R3= R4 = []